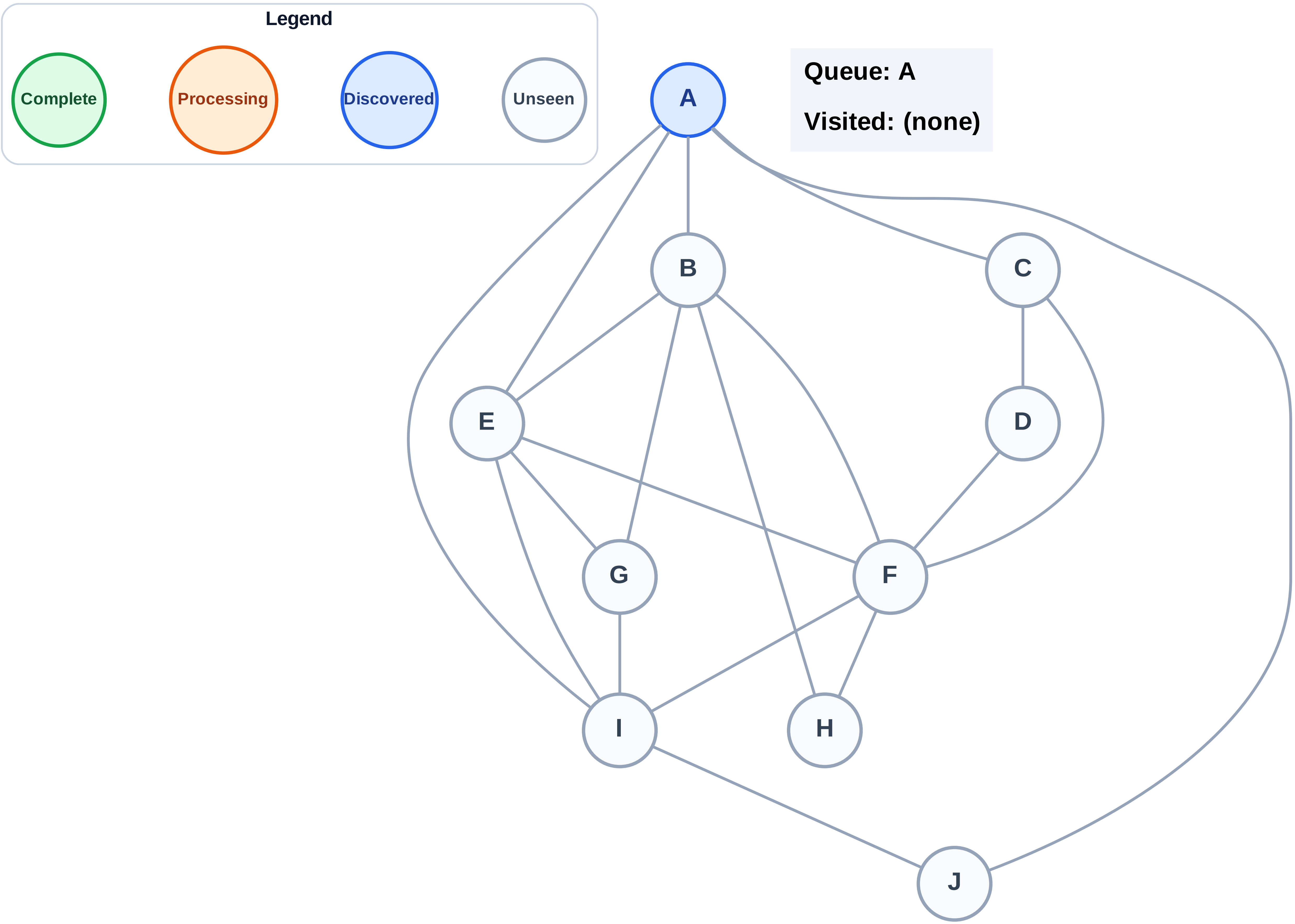
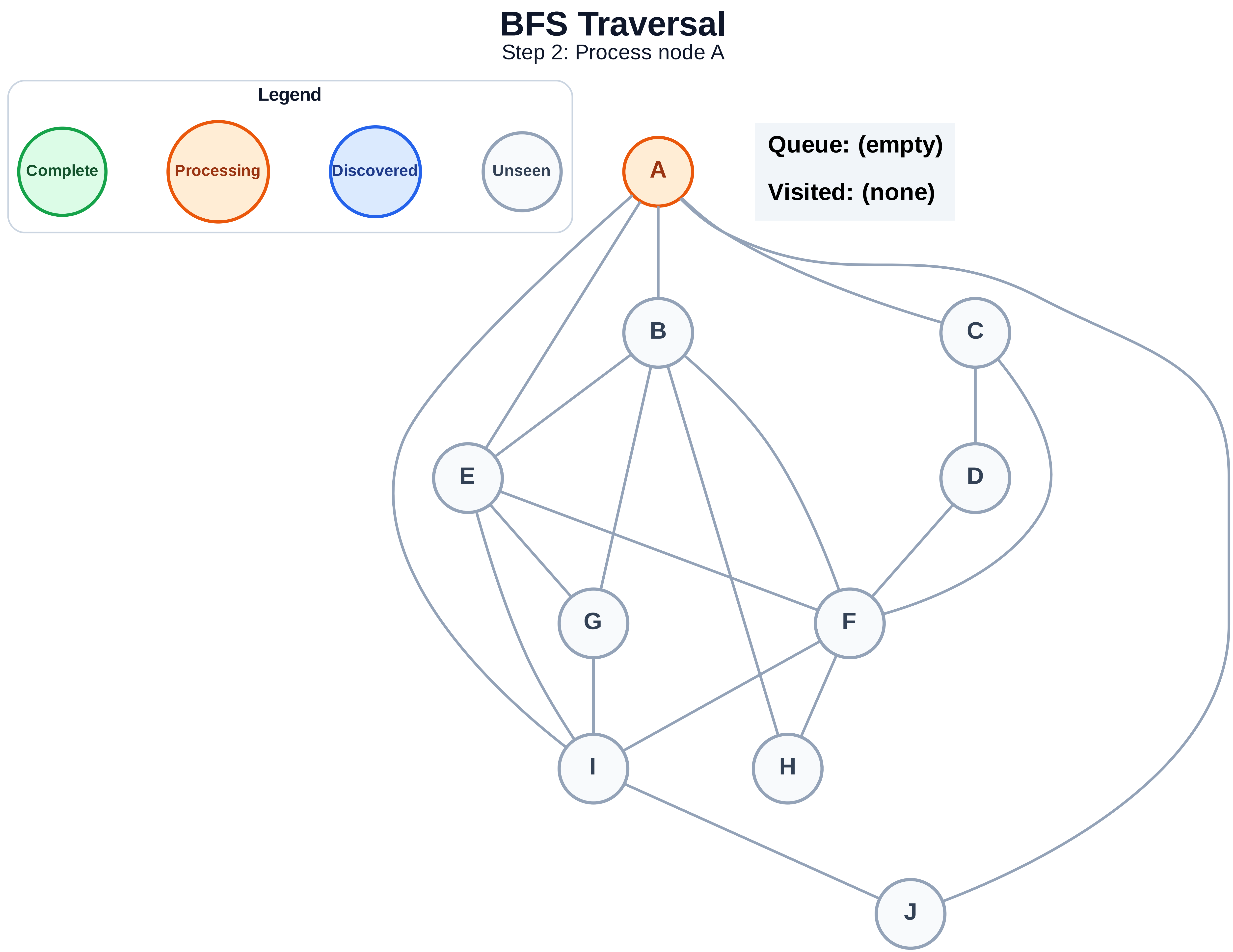
BFS Traversal

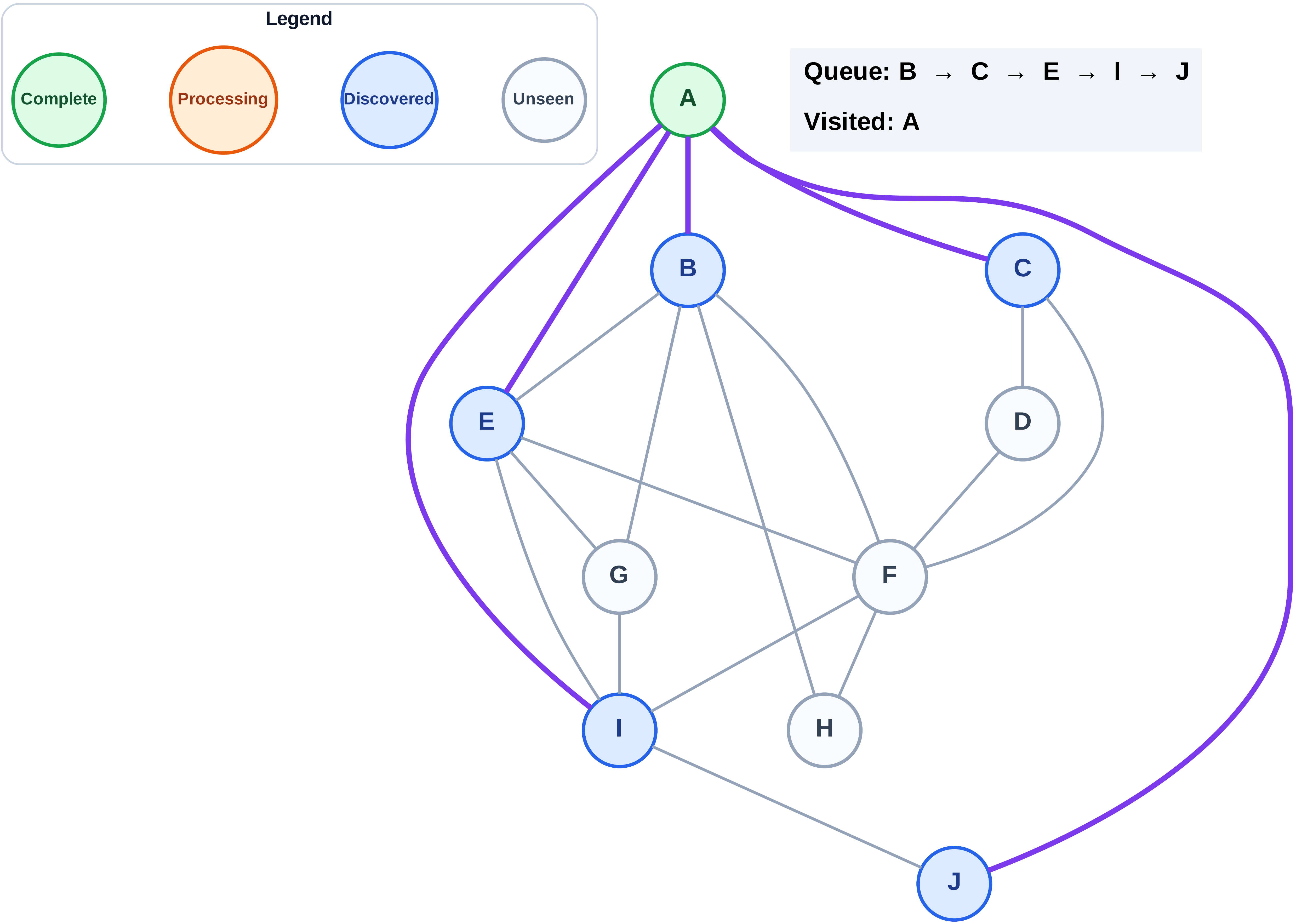
Step 1: Start at A




BFS Traversal
Step 2: Process node A
Legend
Complete
Processing
Discovered
Unseen
Queue: (empty)
Visited: (none)
A
B
C
D
E
F
G
H
I
J

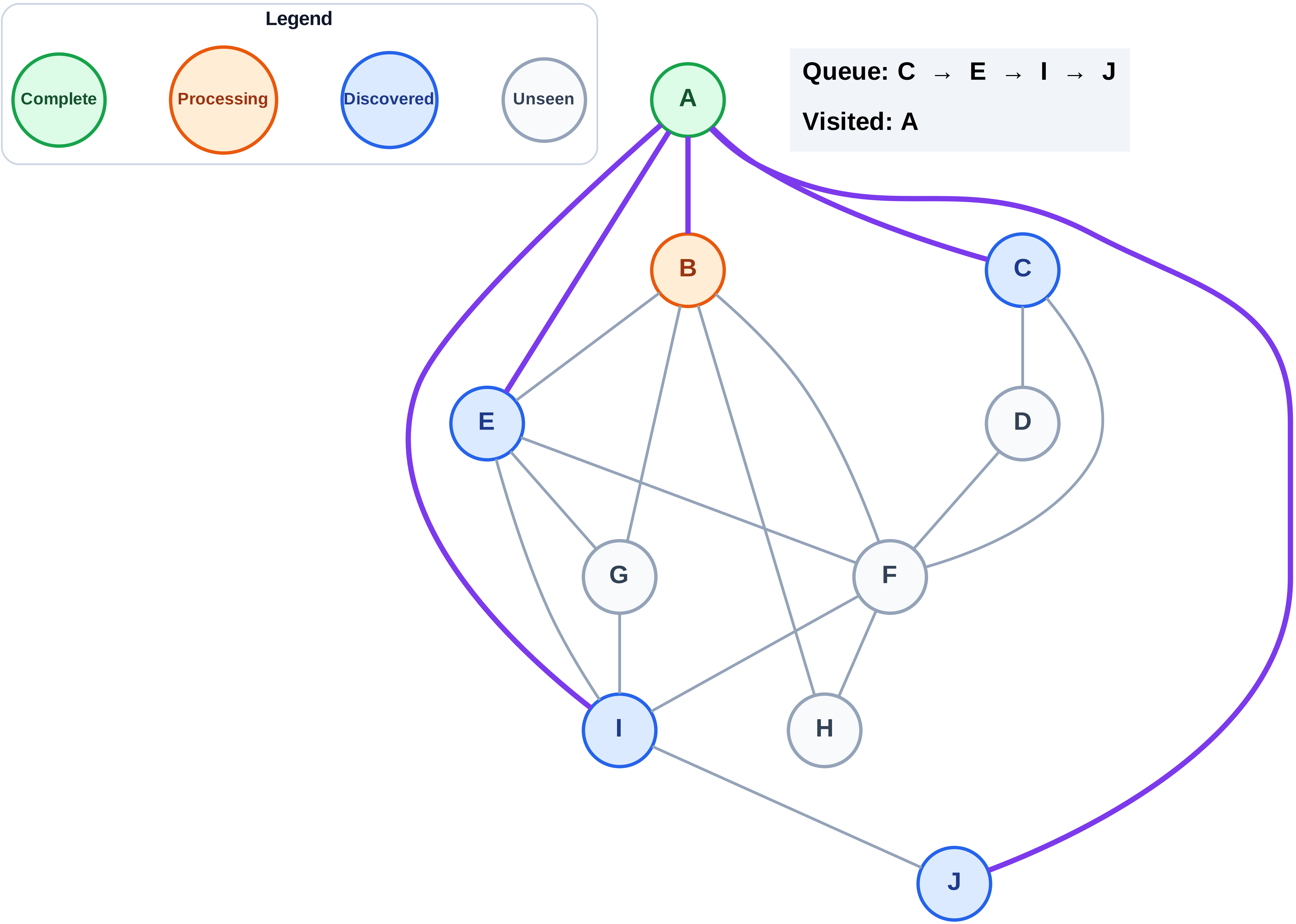
BFS Traversal

Step 3: Discover B, C, E, I, J from A



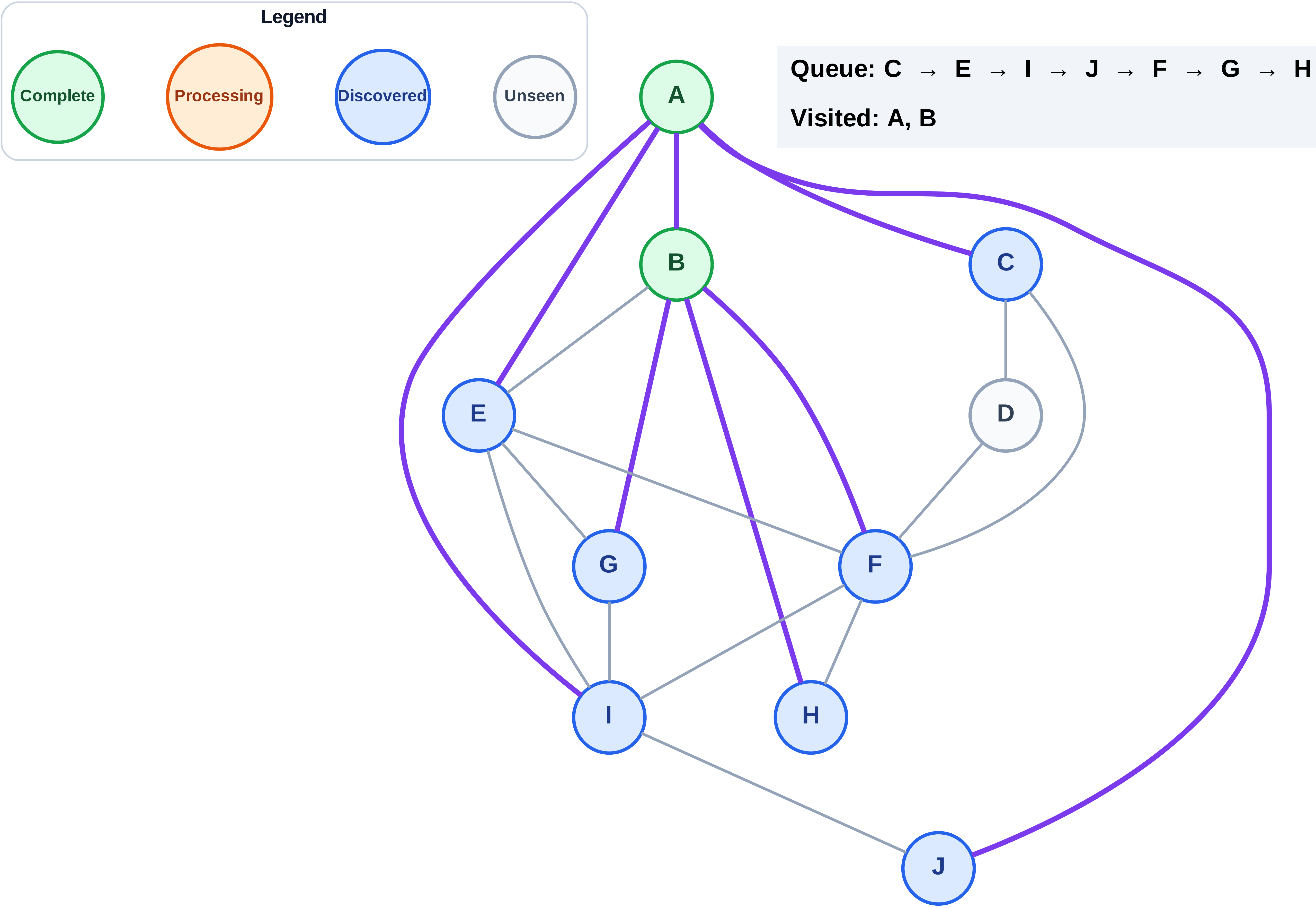
BFS Traversal

Step 4: Process node B



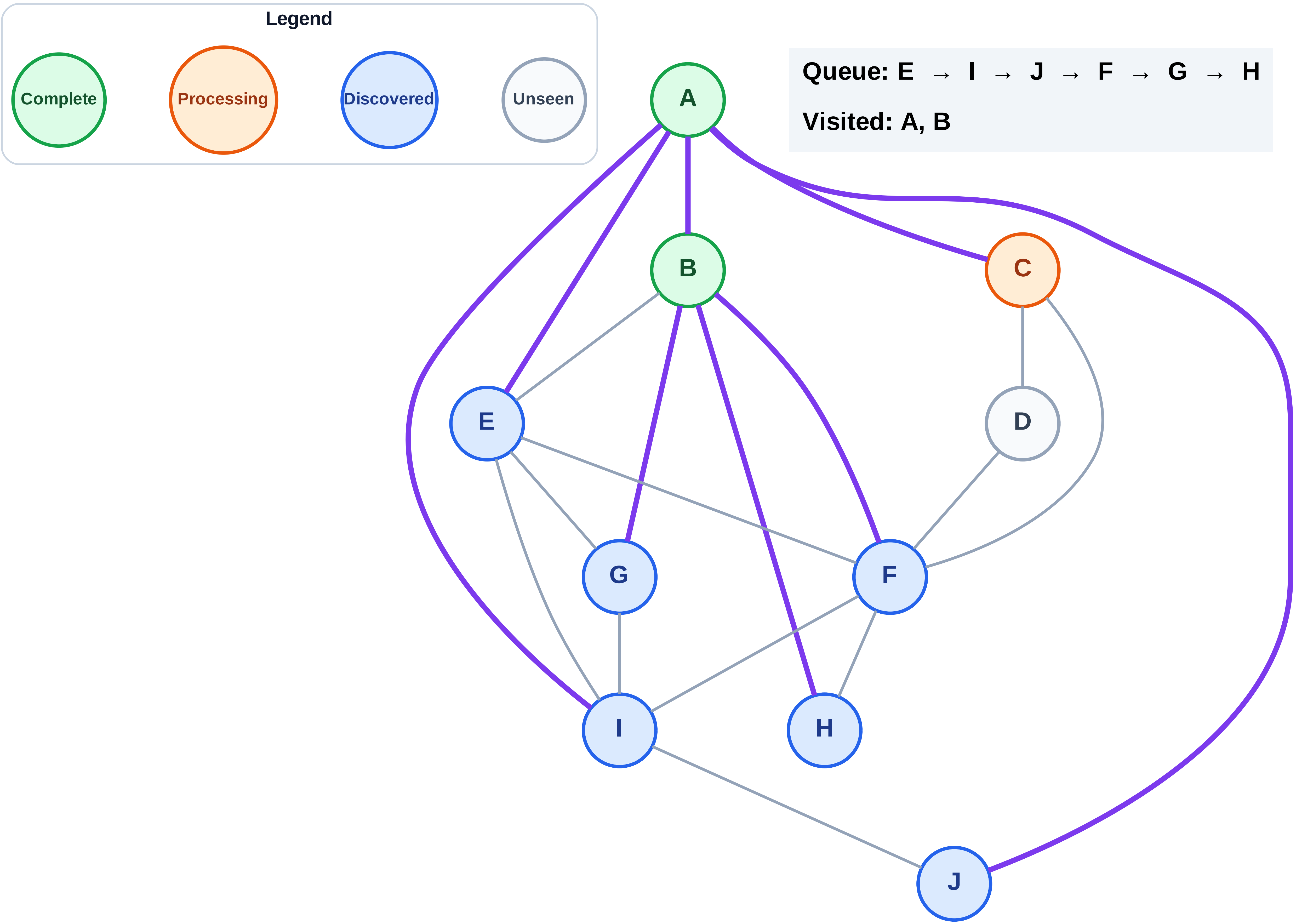
BFS Traversal

Step 5: Discover F, G, H from B



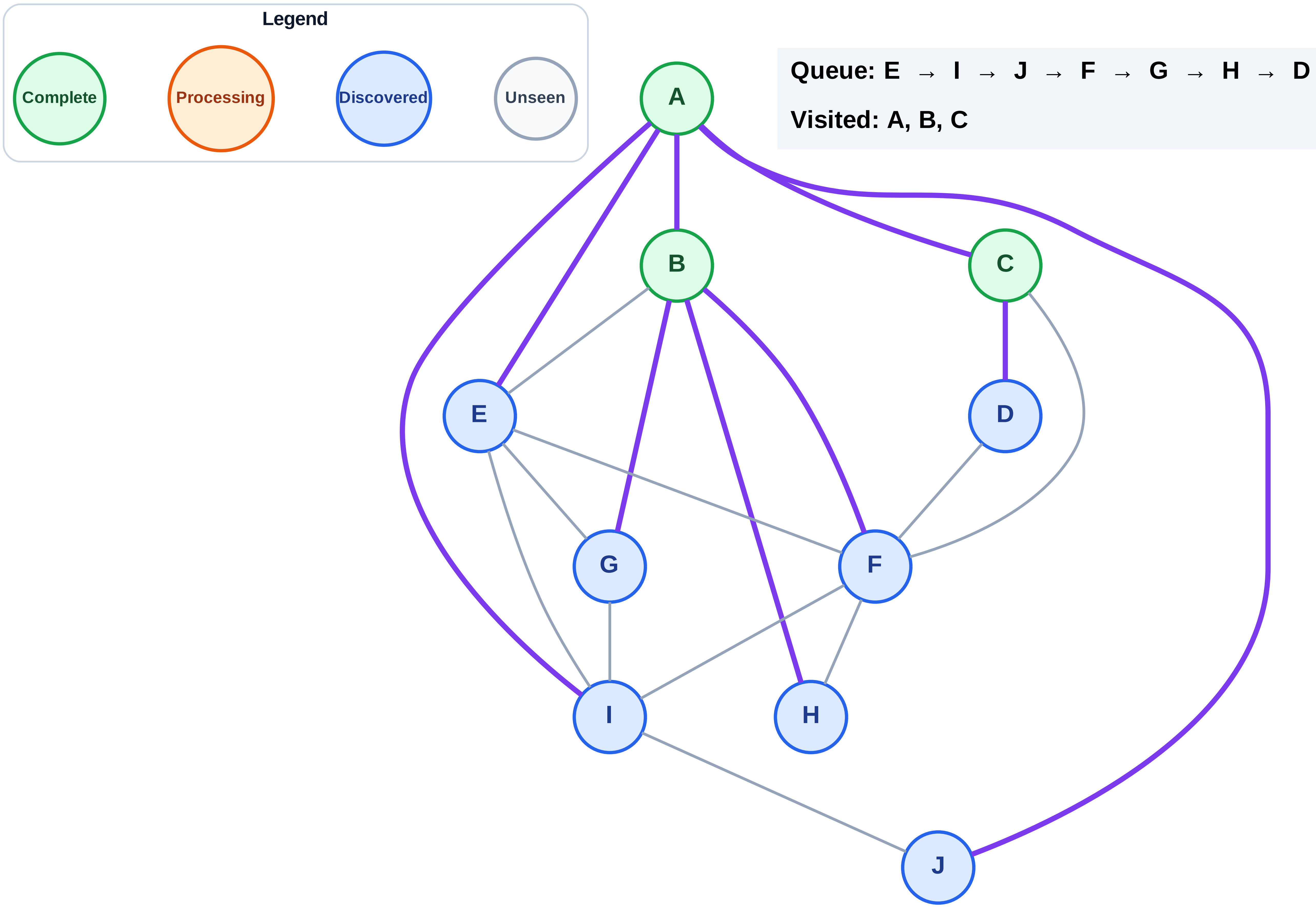
BFS Traversal

Step 6: Process node C



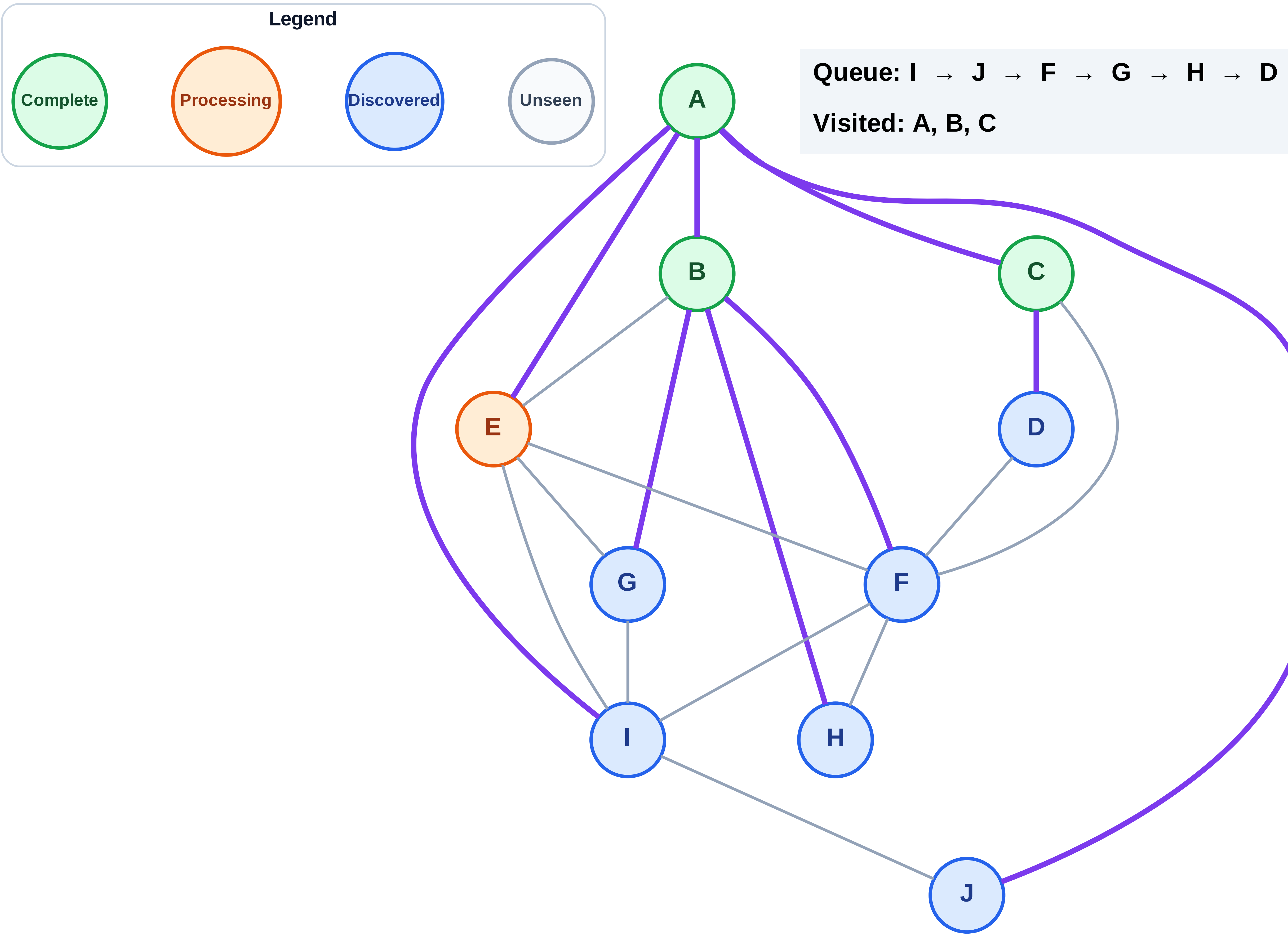
BFS Traversal

Step 7: Discover D from C



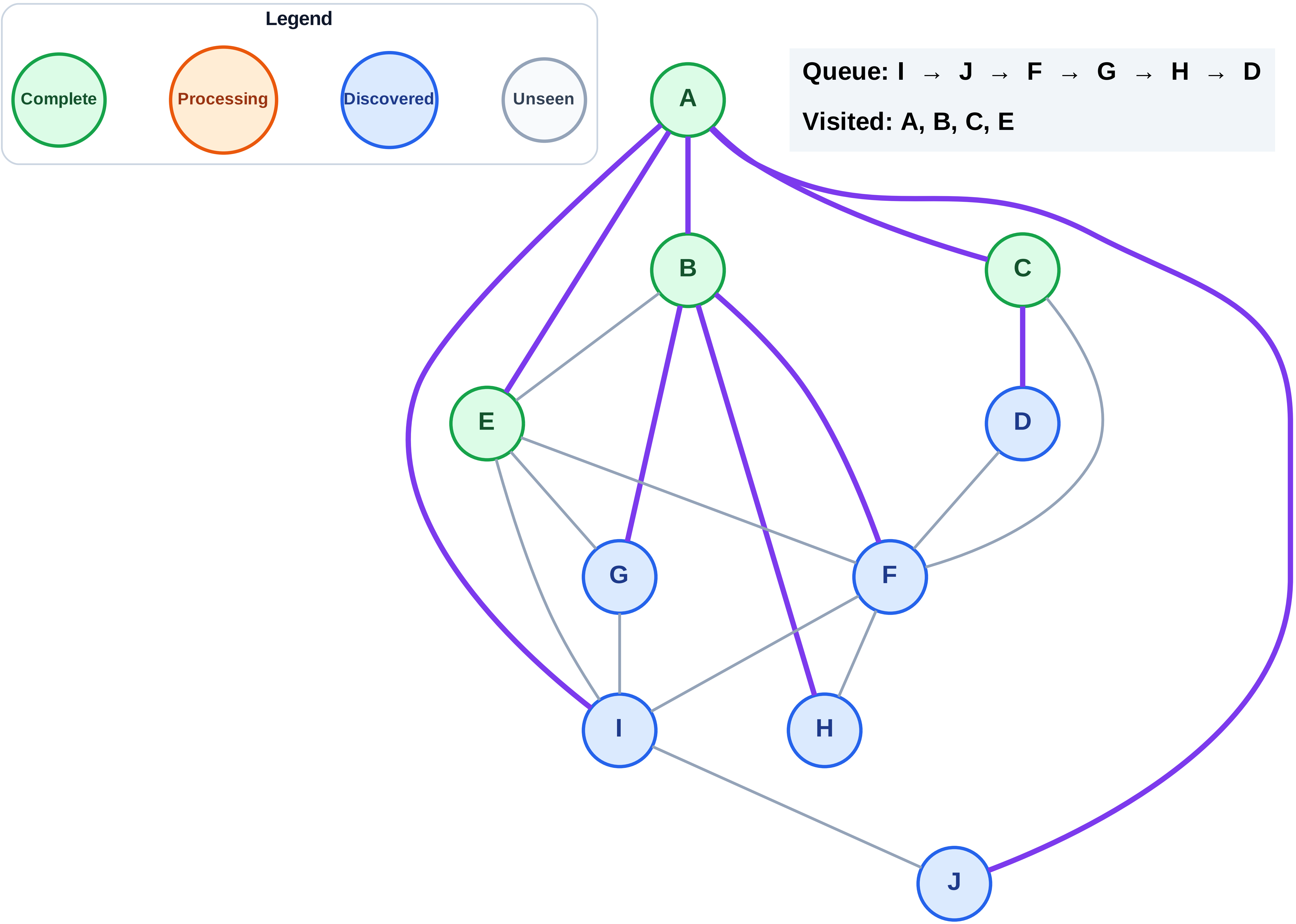
BFS Traversal

Step 8: Process node E



BFS Traversal

Step 9: No new neighbors from E

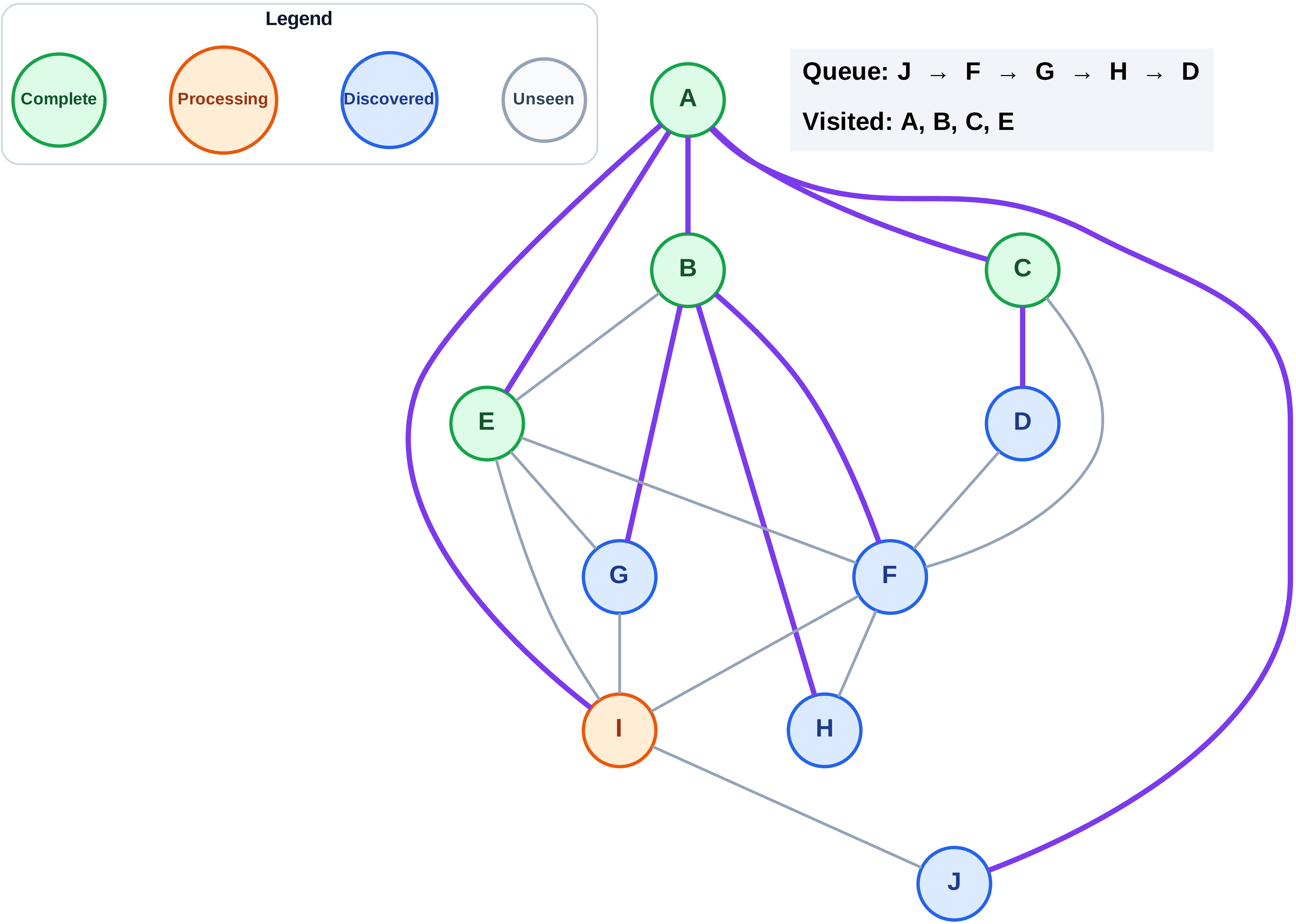


Queue: I → J → F → G → H → D

Visited: A, B, C, E

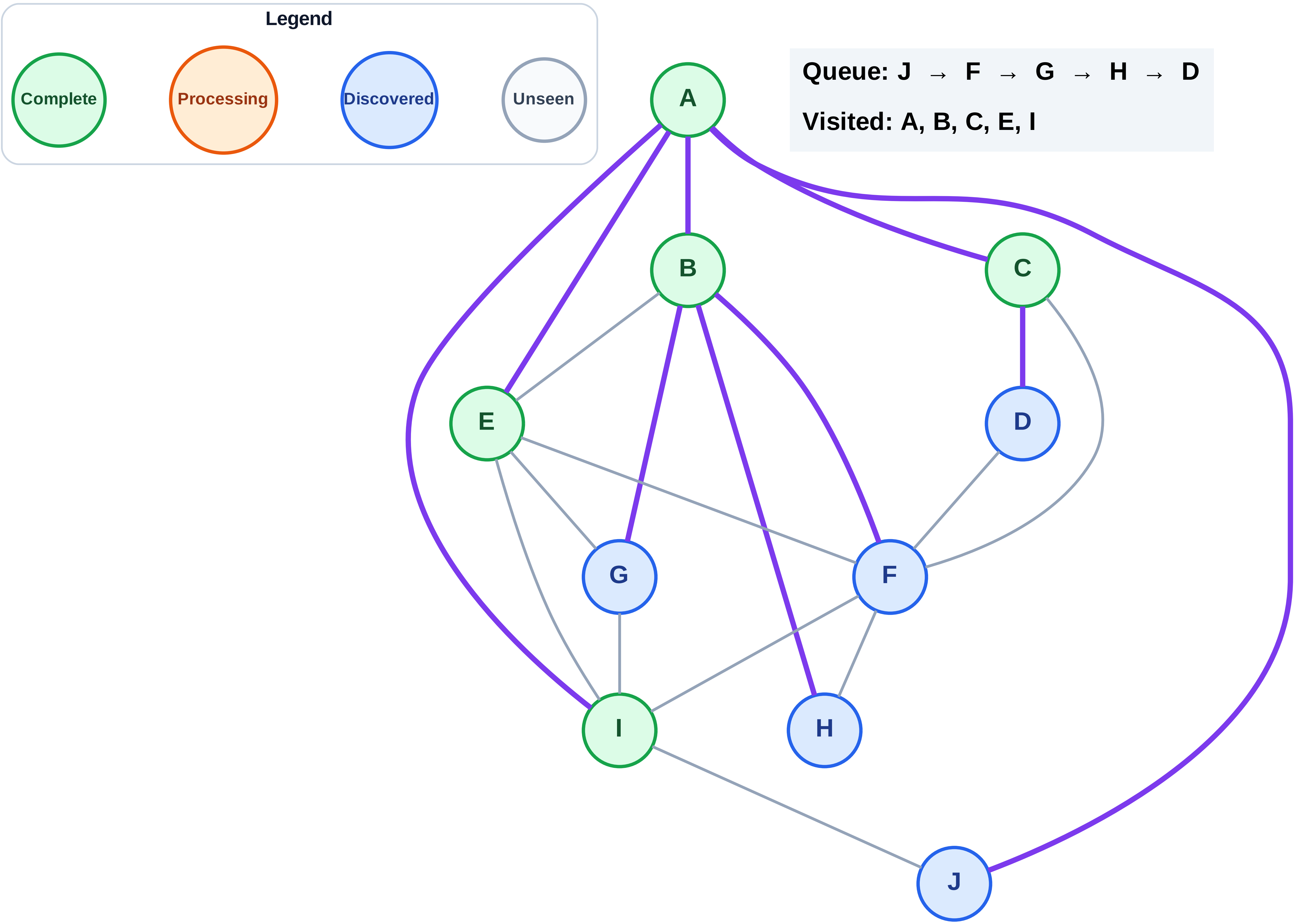
BFS Traversal

Step 10: Process node I



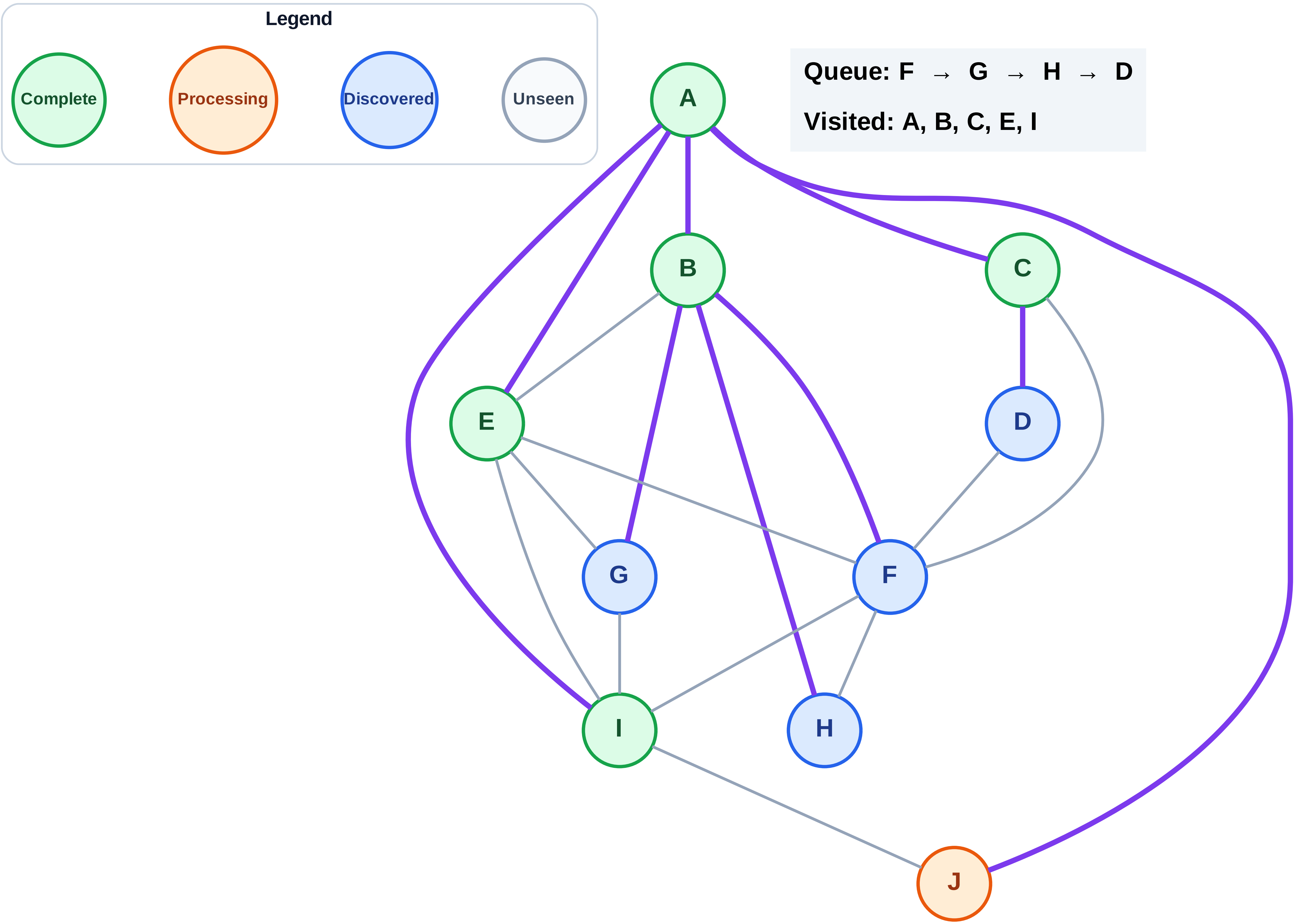
BFS Traversal

Step 11: No new neighbors from I



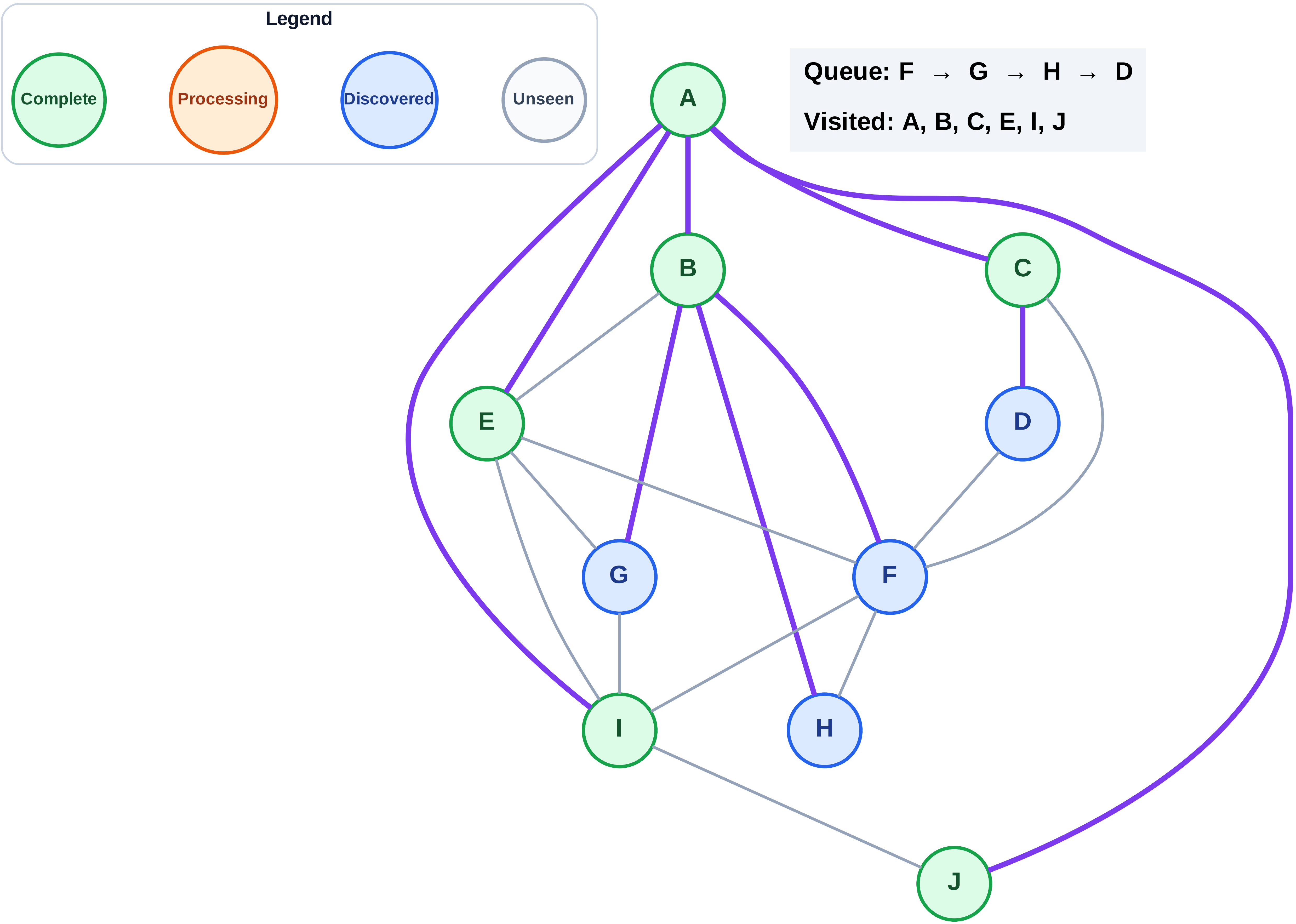
BFS Traversal

Step 12: Process node J



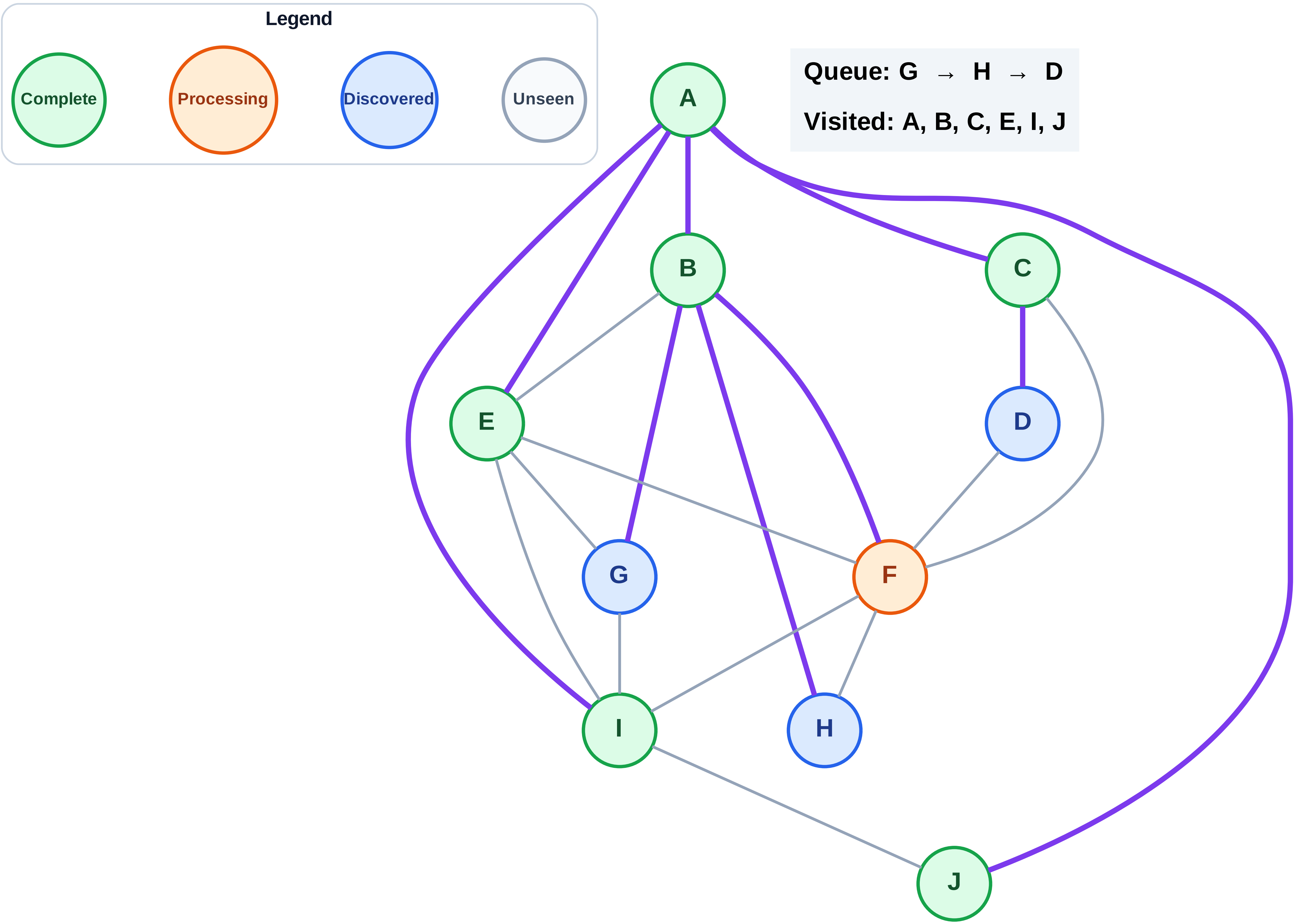
BFS Traversal

Step 13: No new neighbors from J



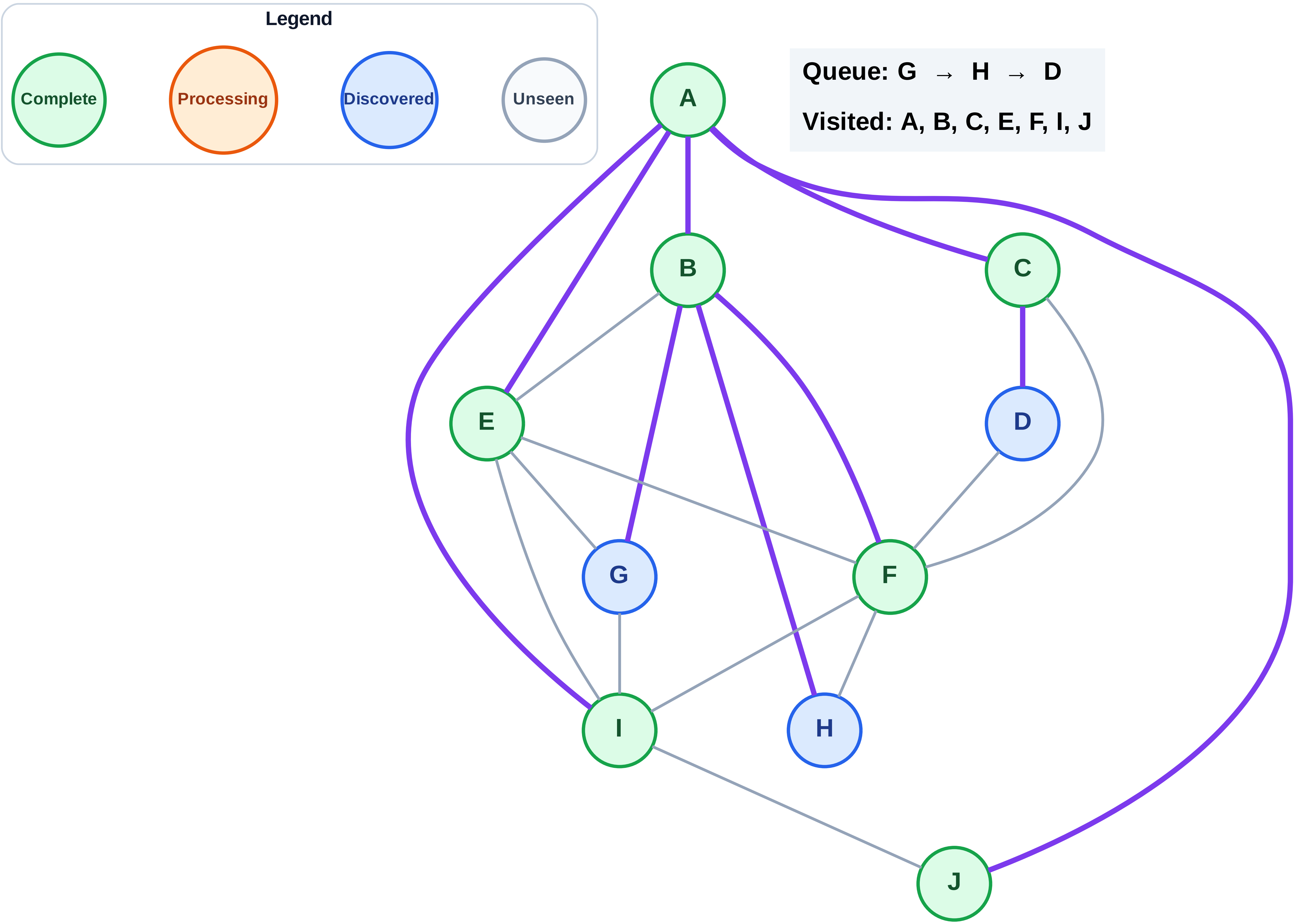
BFS Traversal

Step 14: Process node F



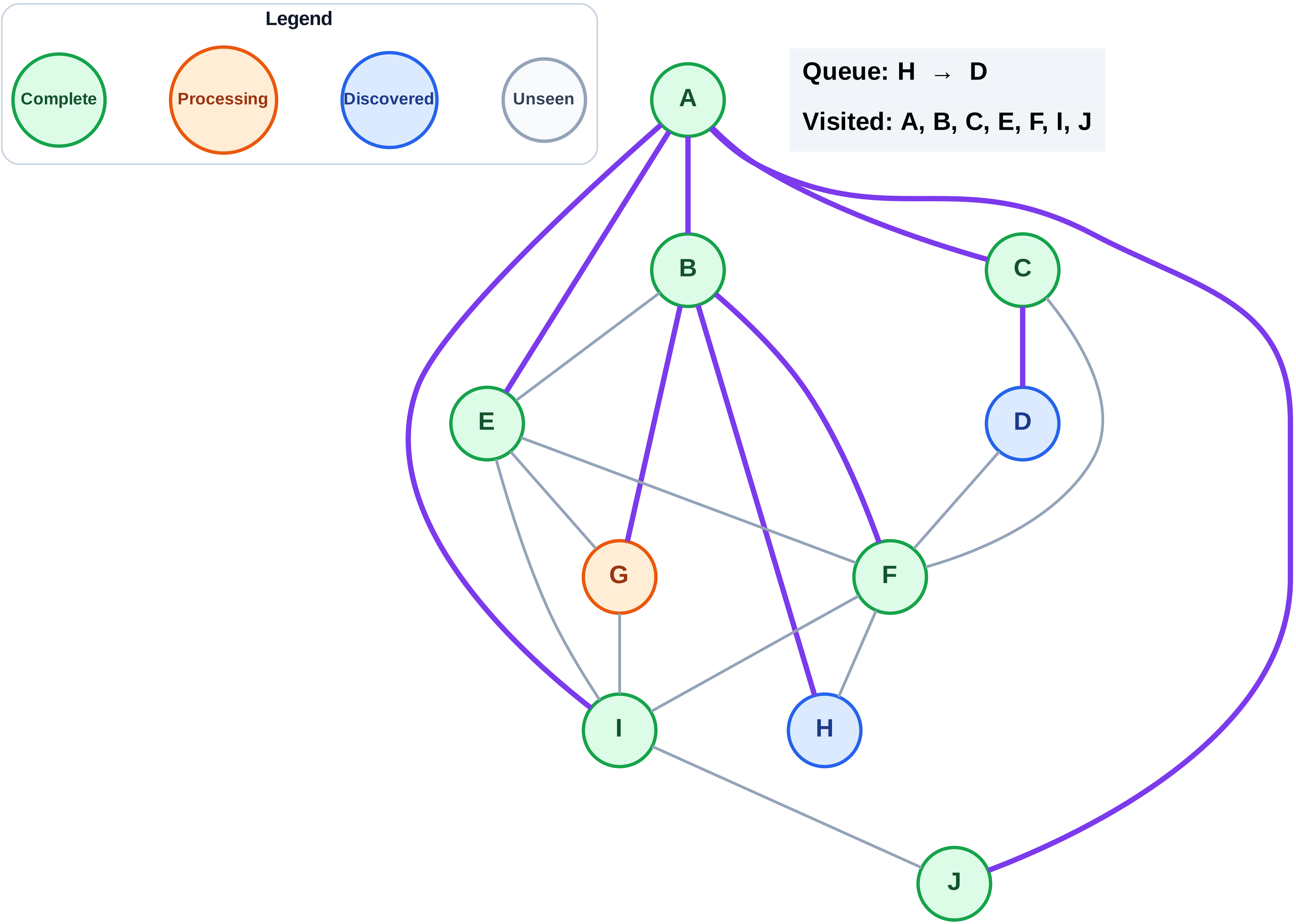
BFS Traversal

Step 15: No new neighbors from F



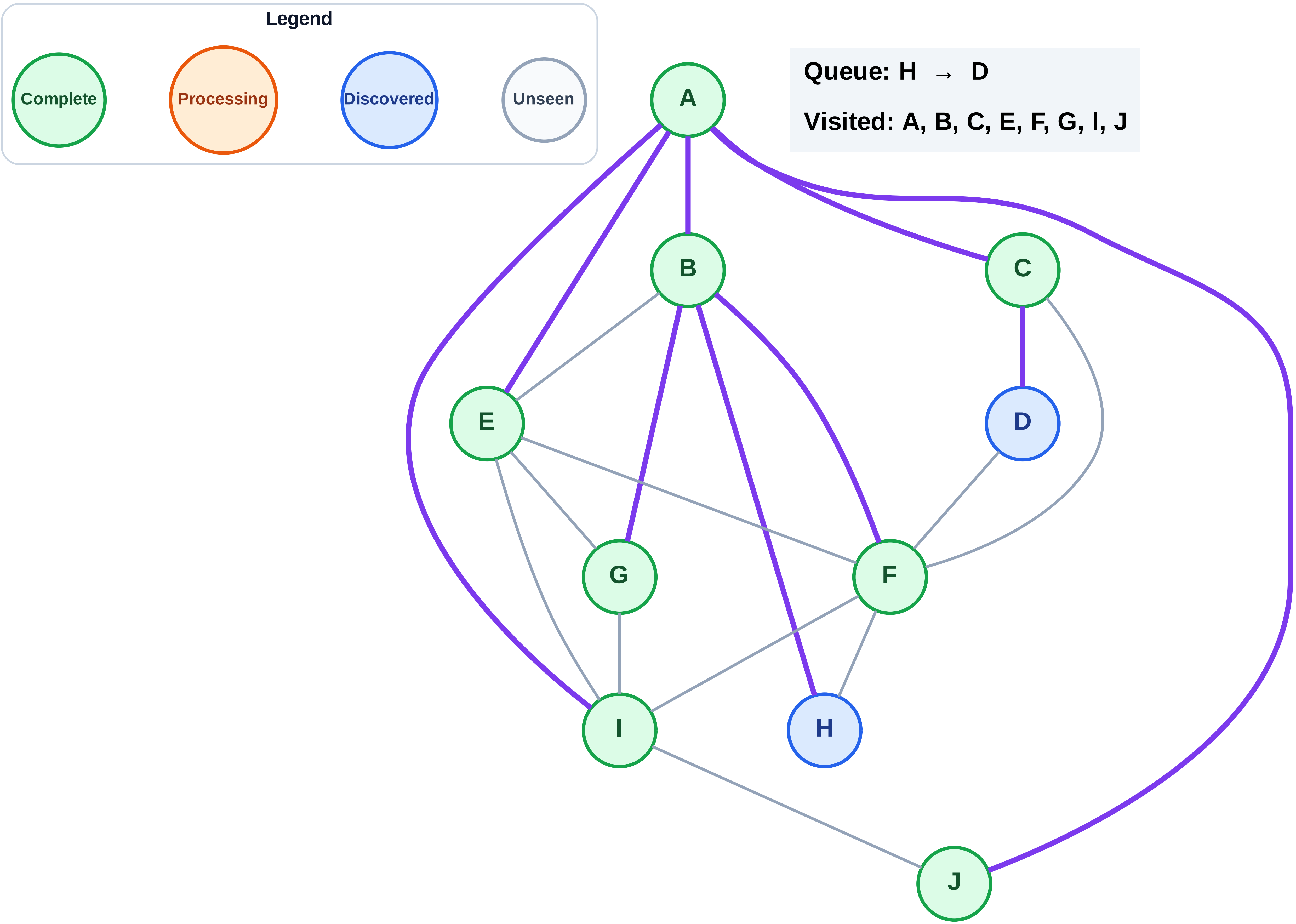
BFS Traversal

Step 16: Process node G



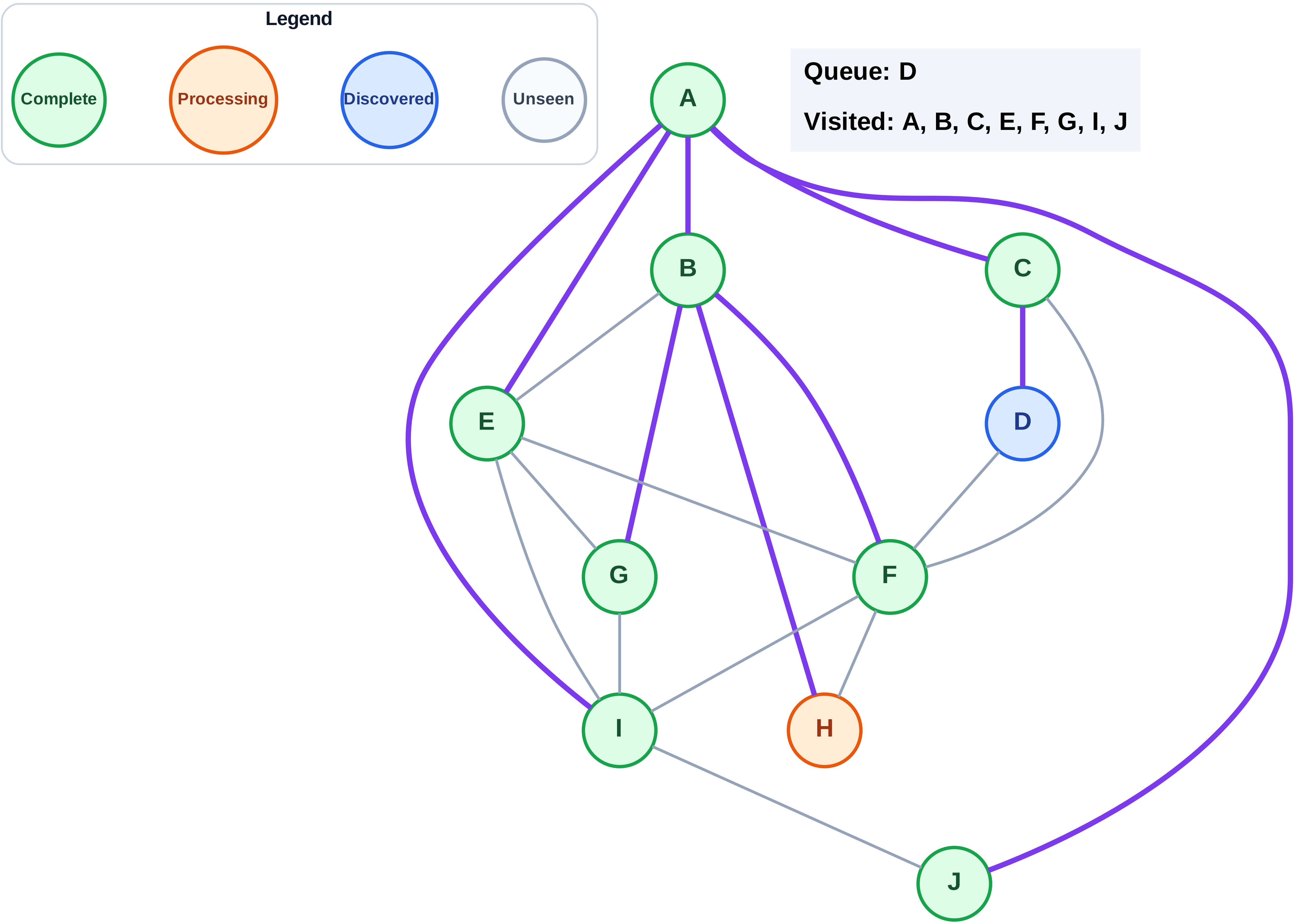
BFS Traversal

Step 17: No new neighbors from G



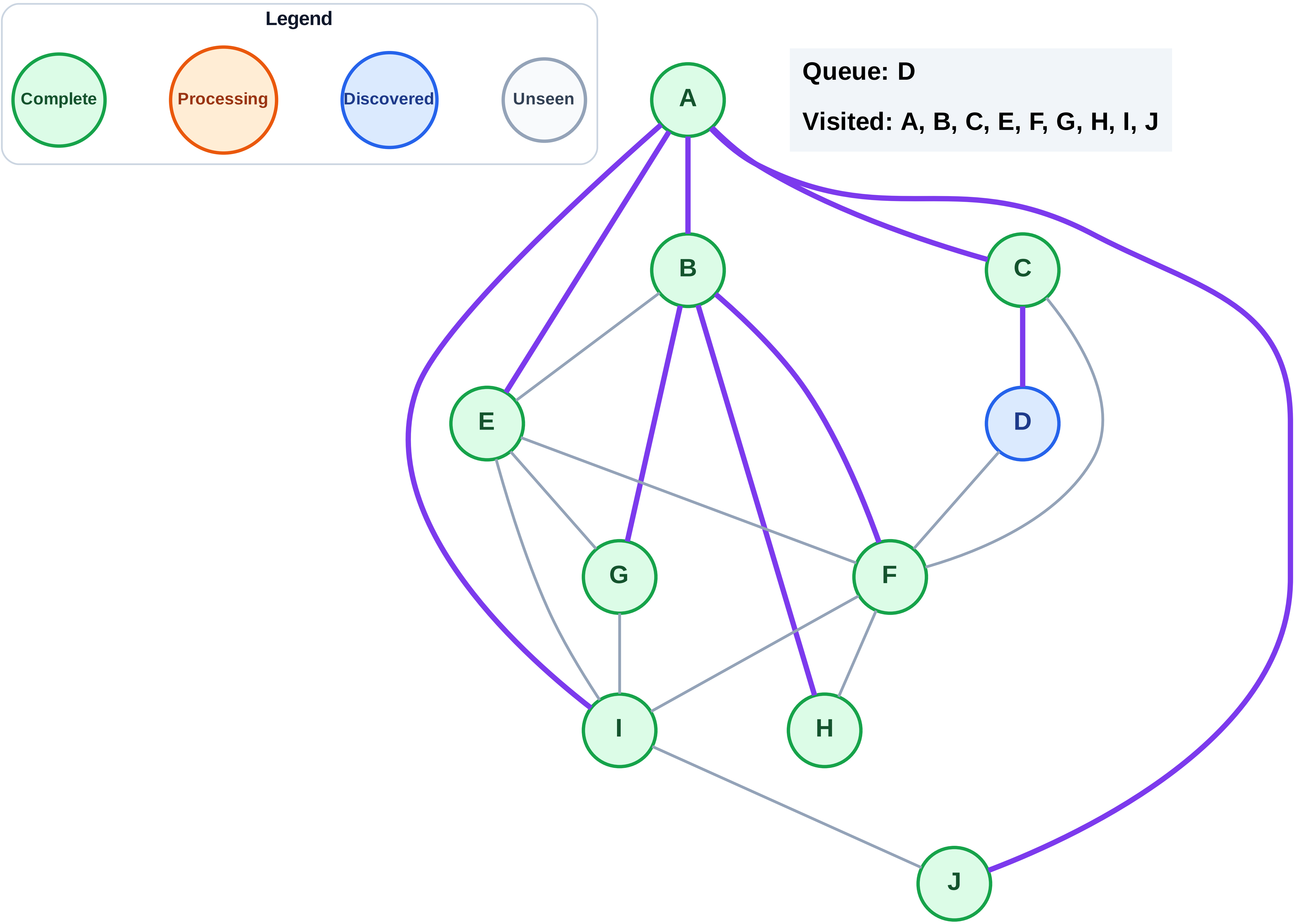
BFS Traversal

Step 18: Process node H



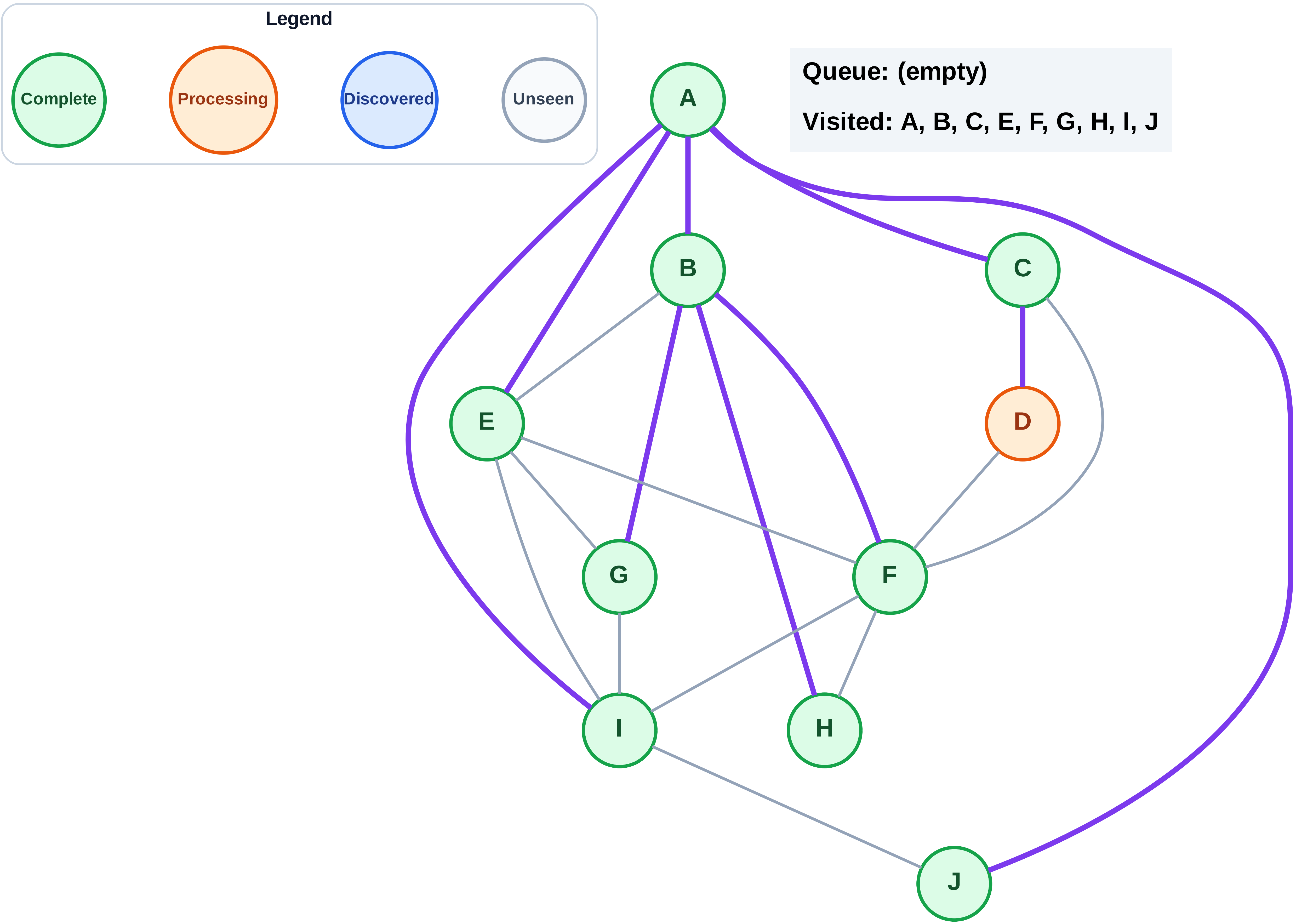
BFS Traversal

Step 19: No new neighbors from H



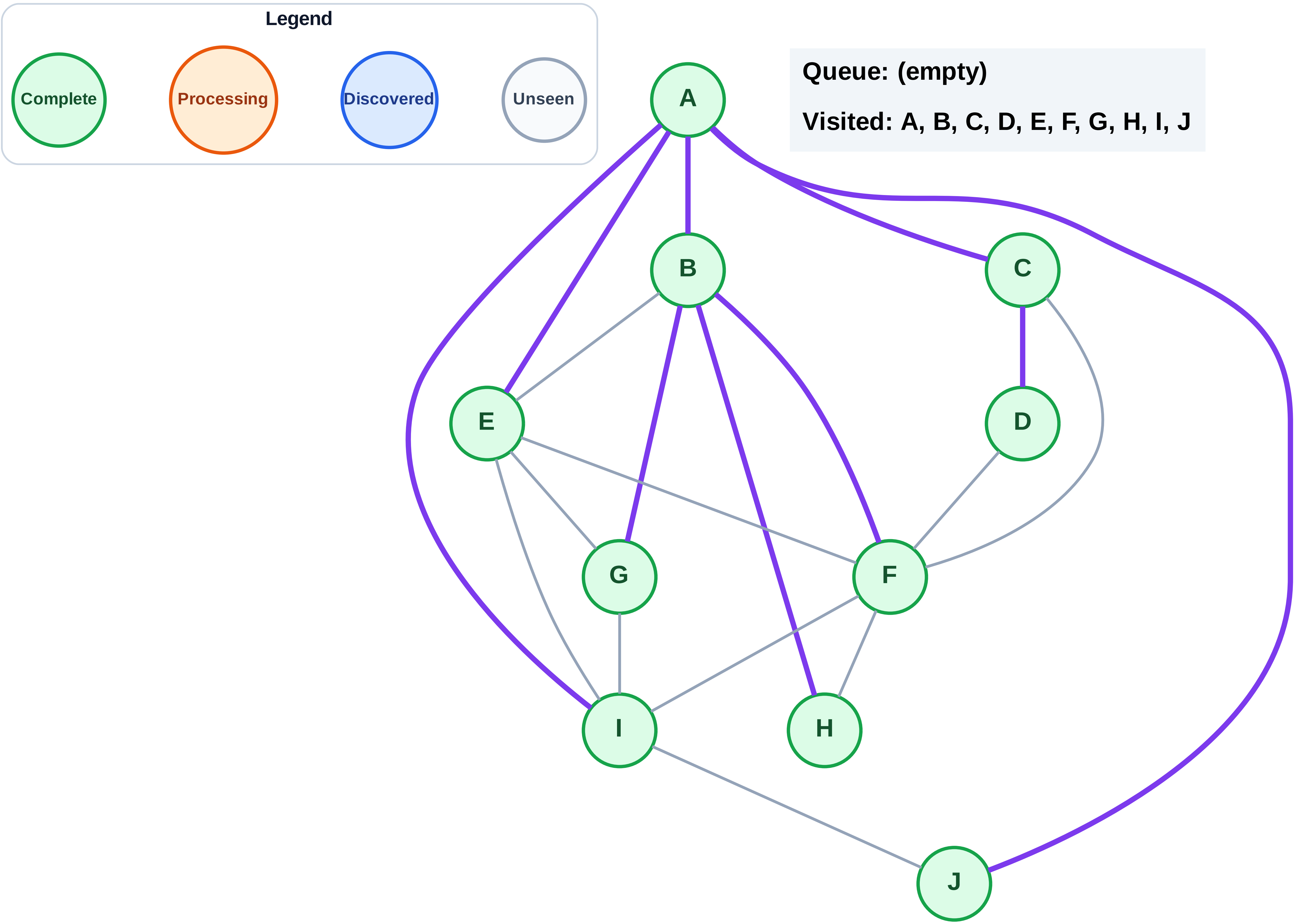
BFS Traversal

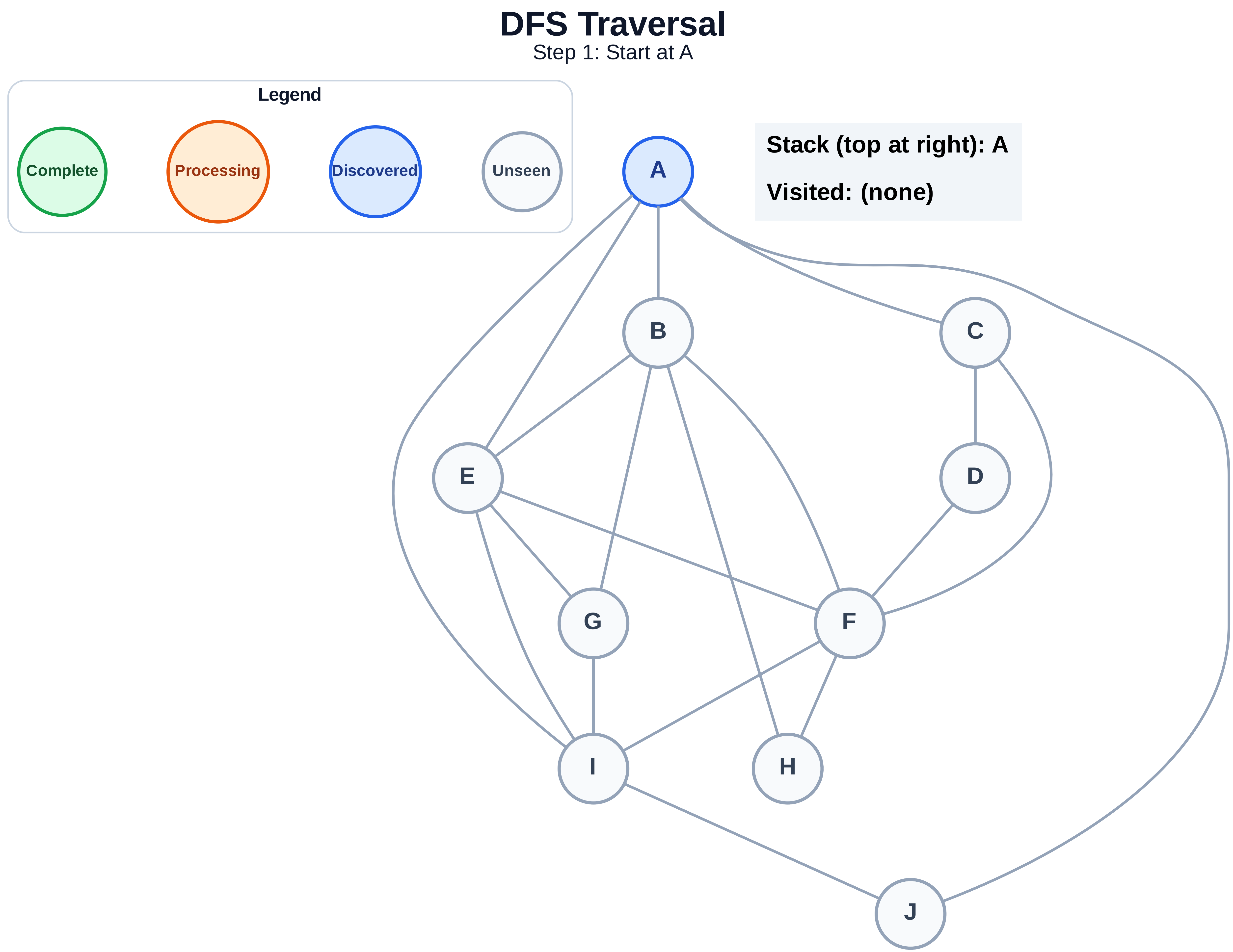
Step 20: Process node D



BFS Traversal

Step 21: No new neighbors from D

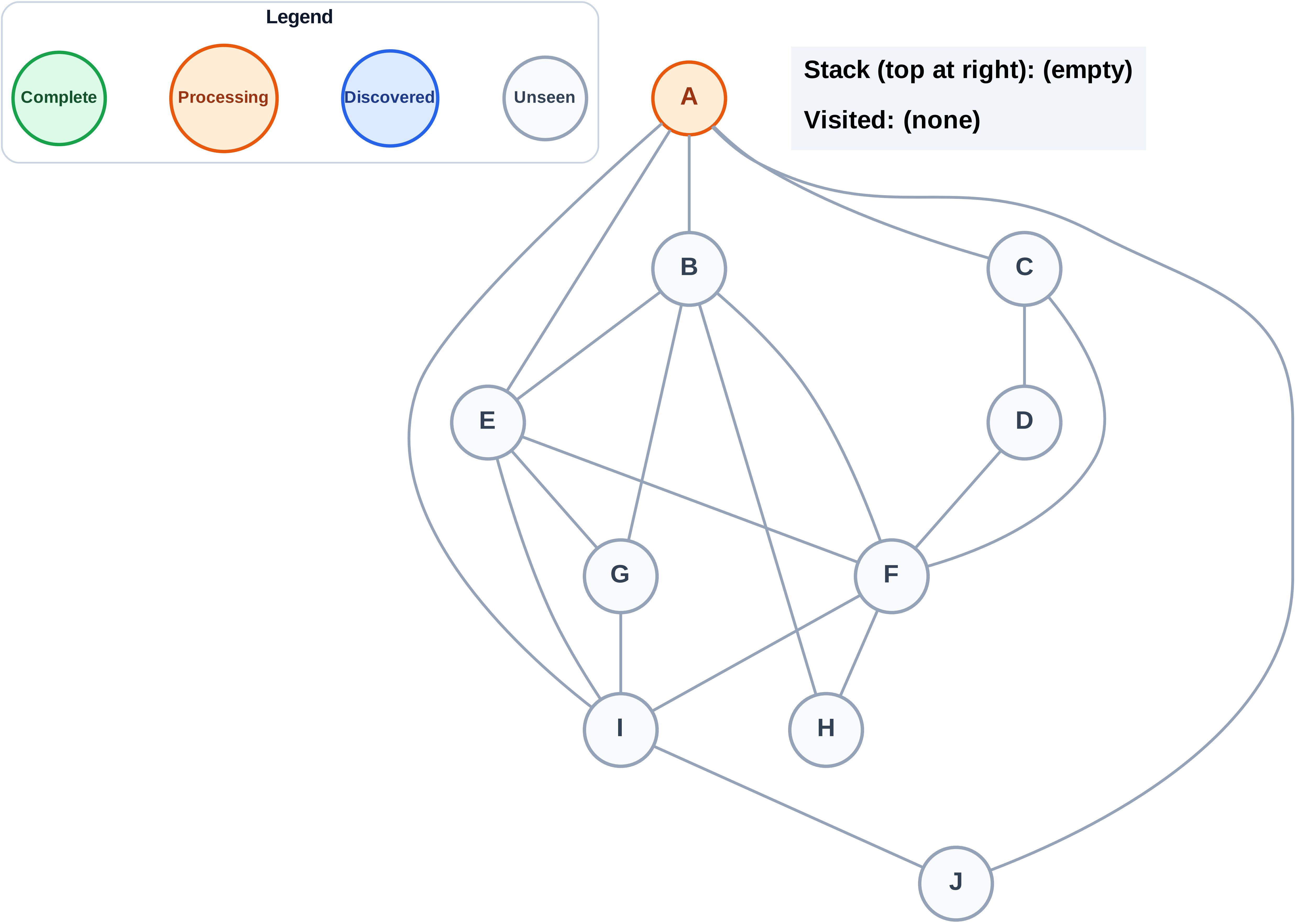




DFS Traversal
Step 1: Start at A
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): A
Visited: (none)
A
B
C
D
E
F
G
H
I
J

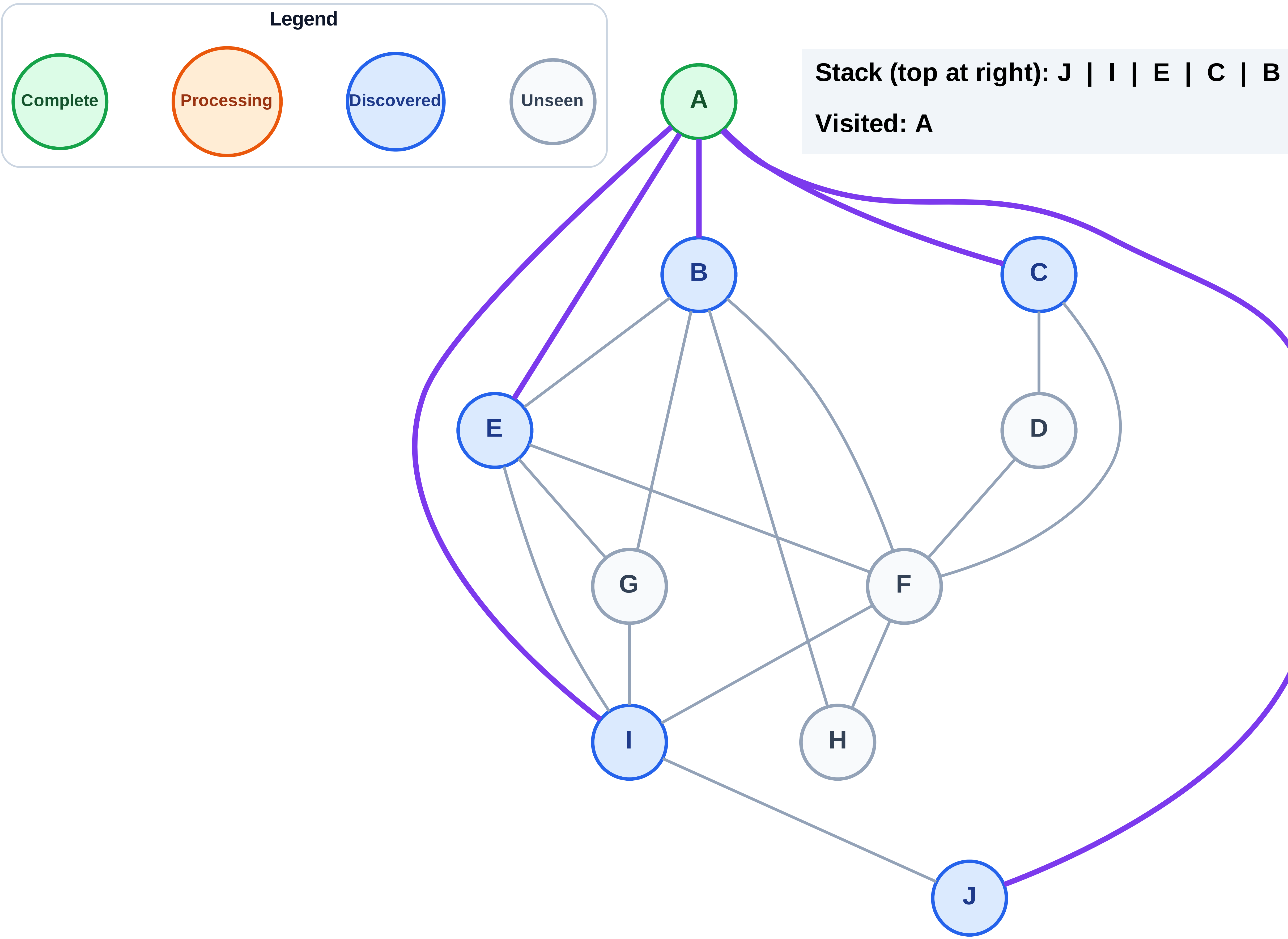
DFS Traversal

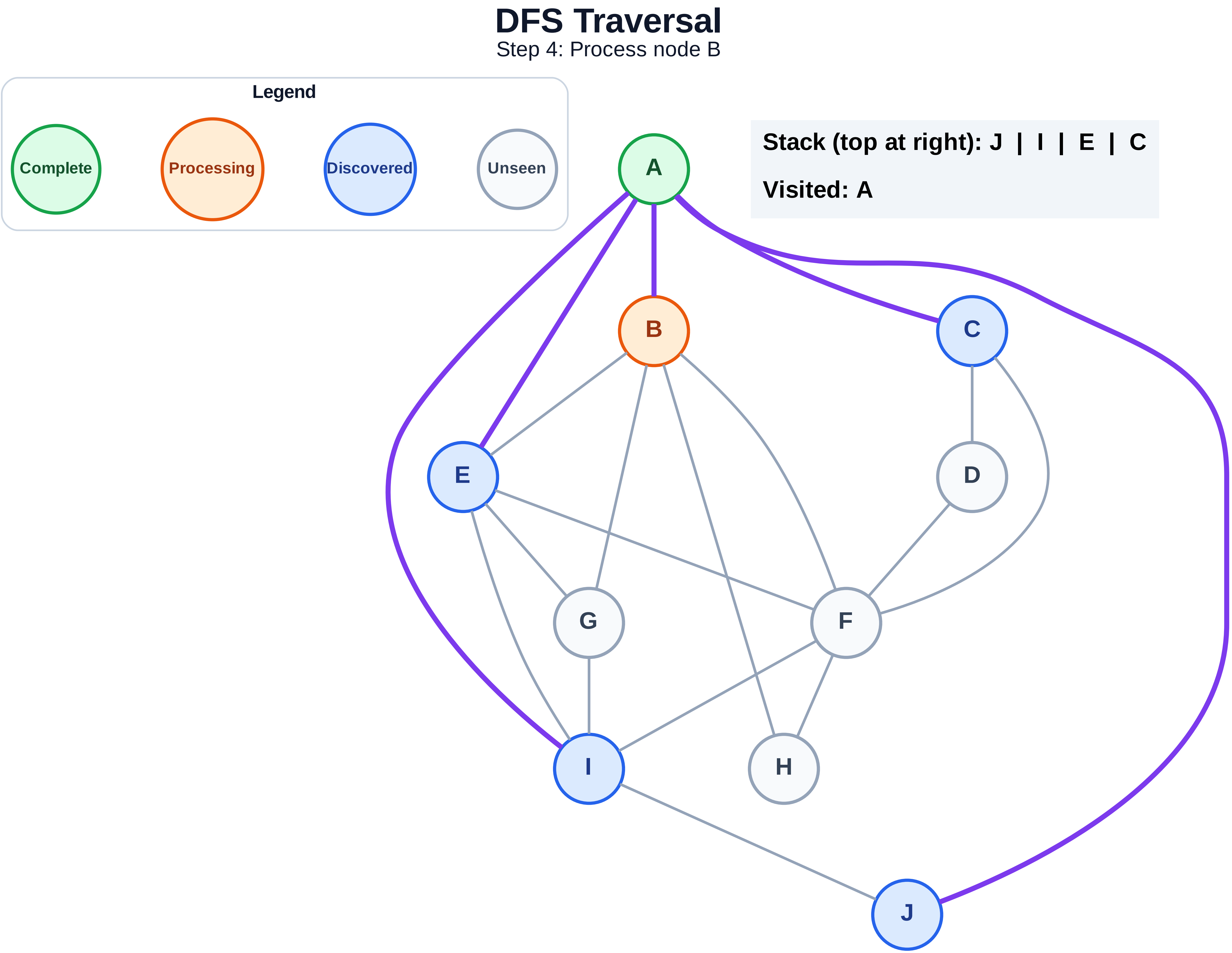
Step 2: Process node A



DFS Traversal

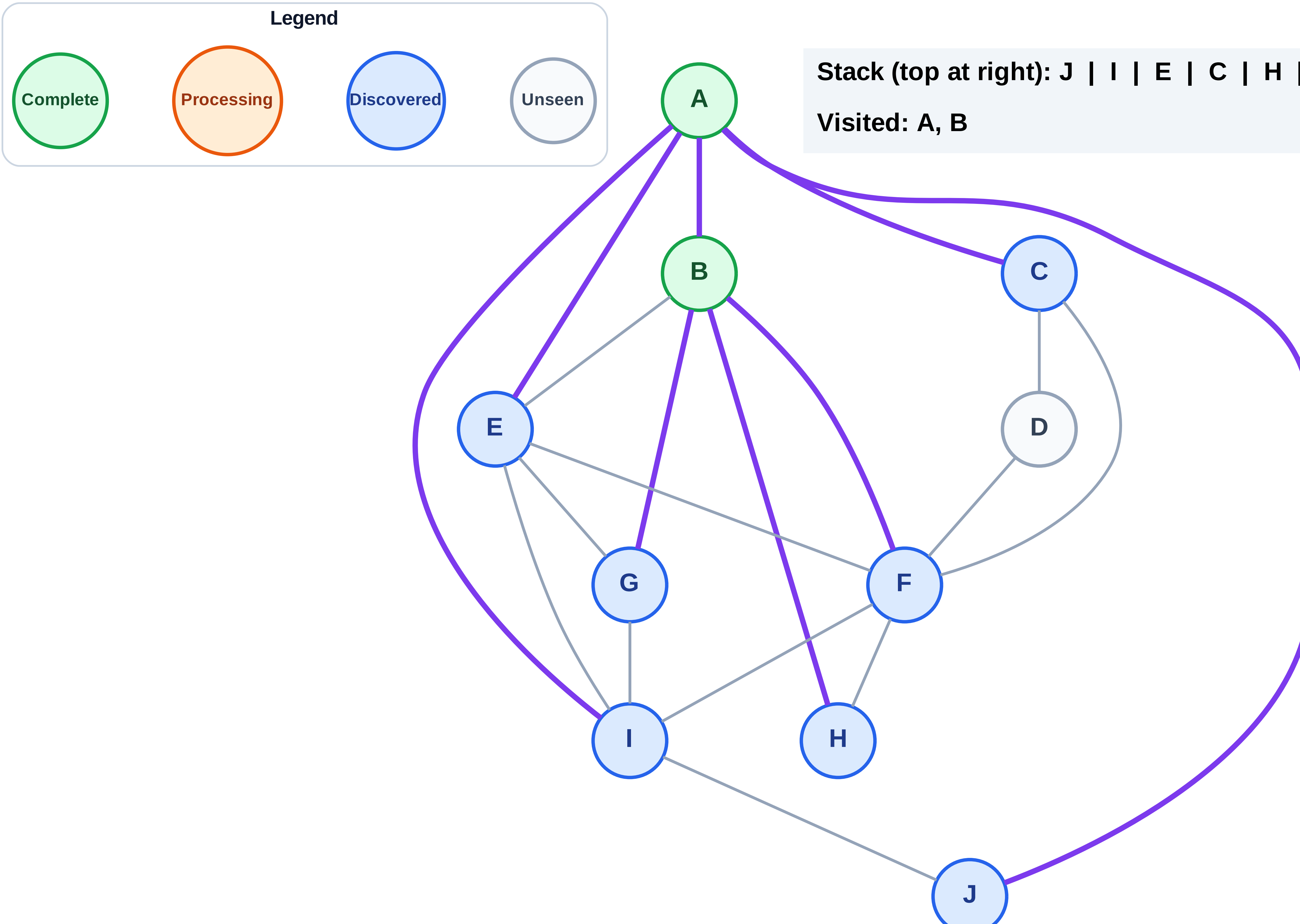
Step 3: Discover J, I, E, C, B from A




DFS Traversal
Step 4: Process node B
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): J | I | E | C
Visited: A
A
B
C
D
E
F
G
H
I
J

DFS Traversal

Step 5: Discover H, G, F from B

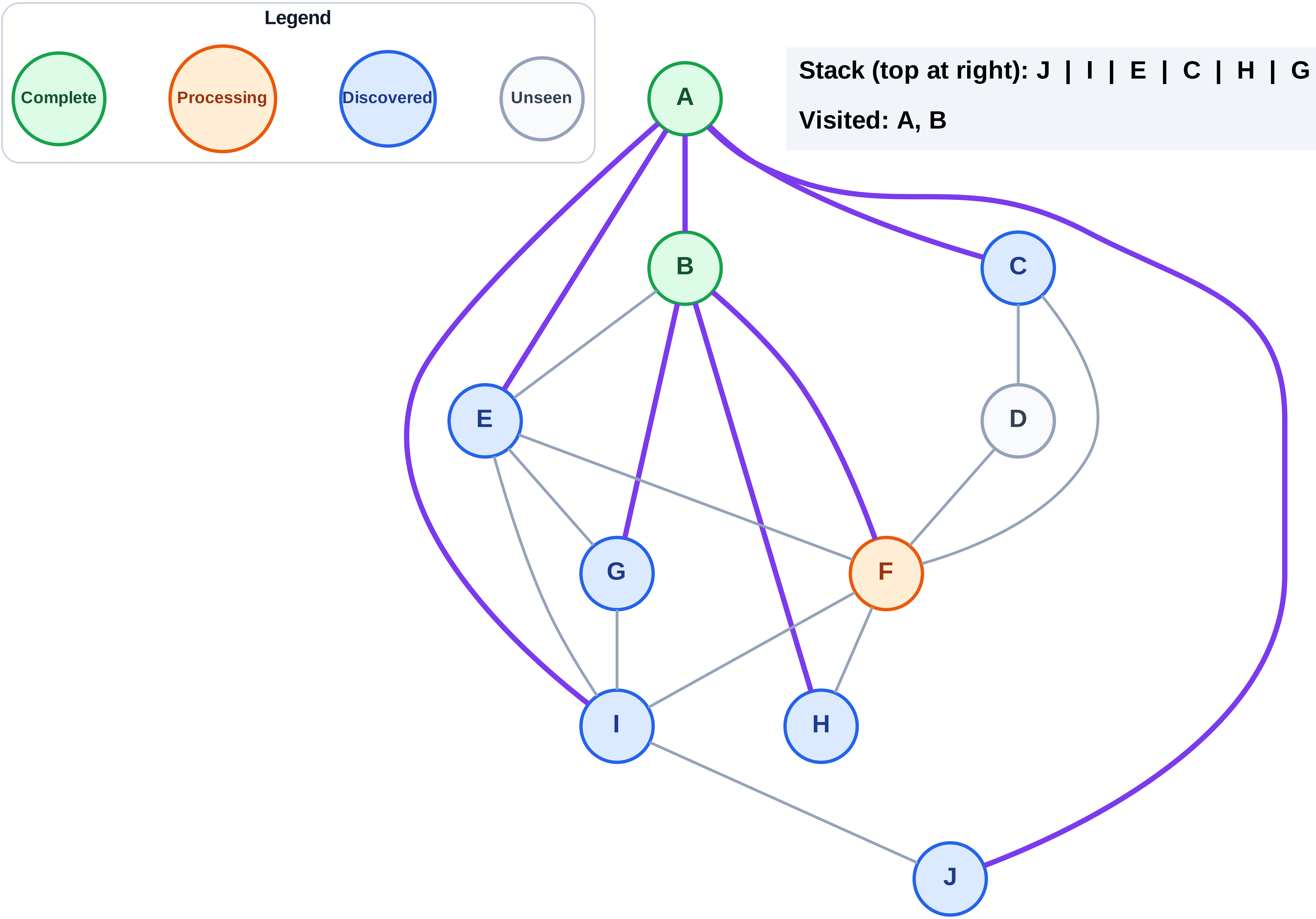


Stack (top at right): J | I | E | C | H | G | F

Visited: A, B

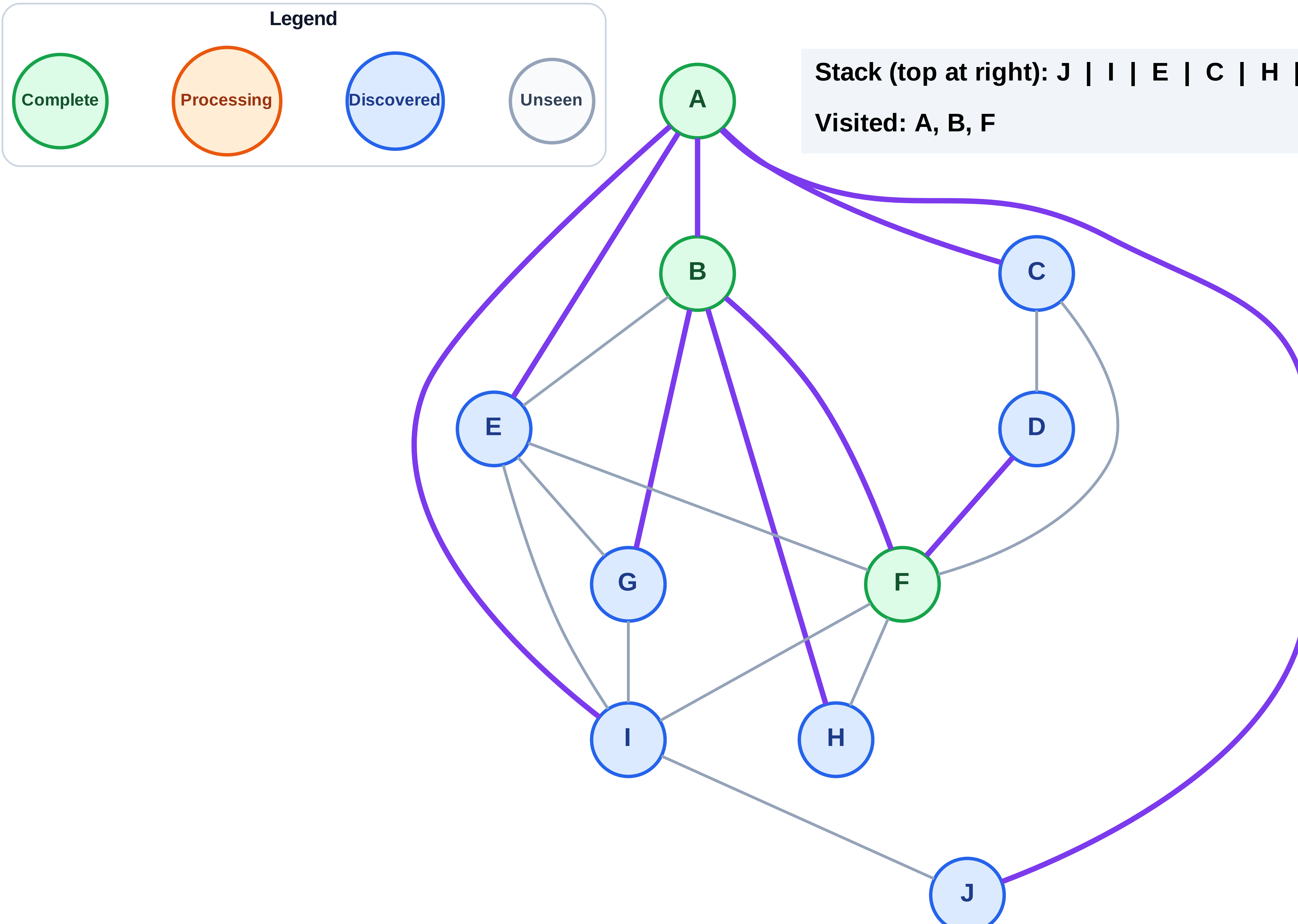
DFS Traversal

Step 6: Process node F



DFS Traversal

Step 7: Discover D from F

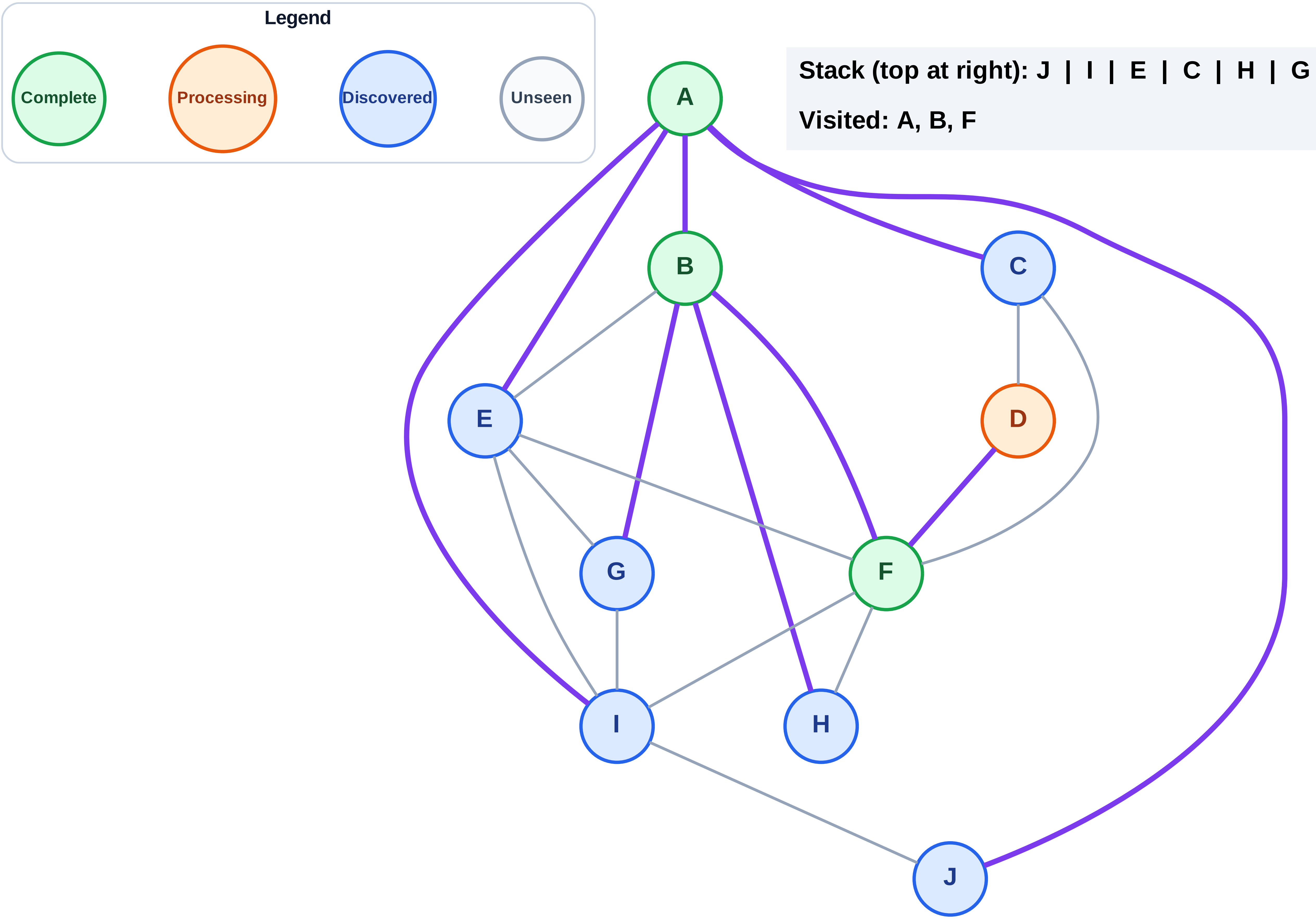


Stack (top at right): J | I | E | C | H | G | D

Visited: A, B, F

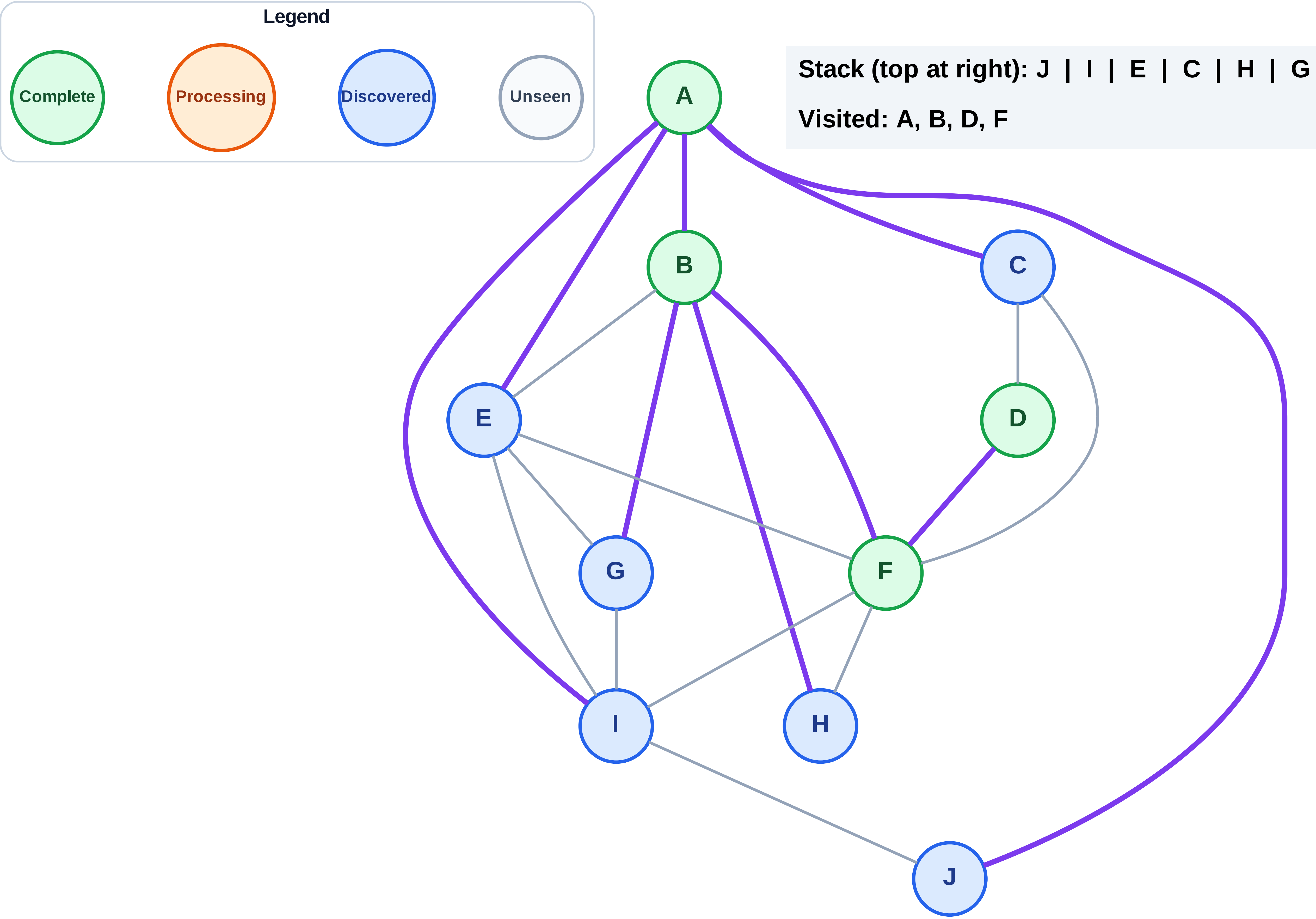
DFS Traversal

Step 8: Process node D



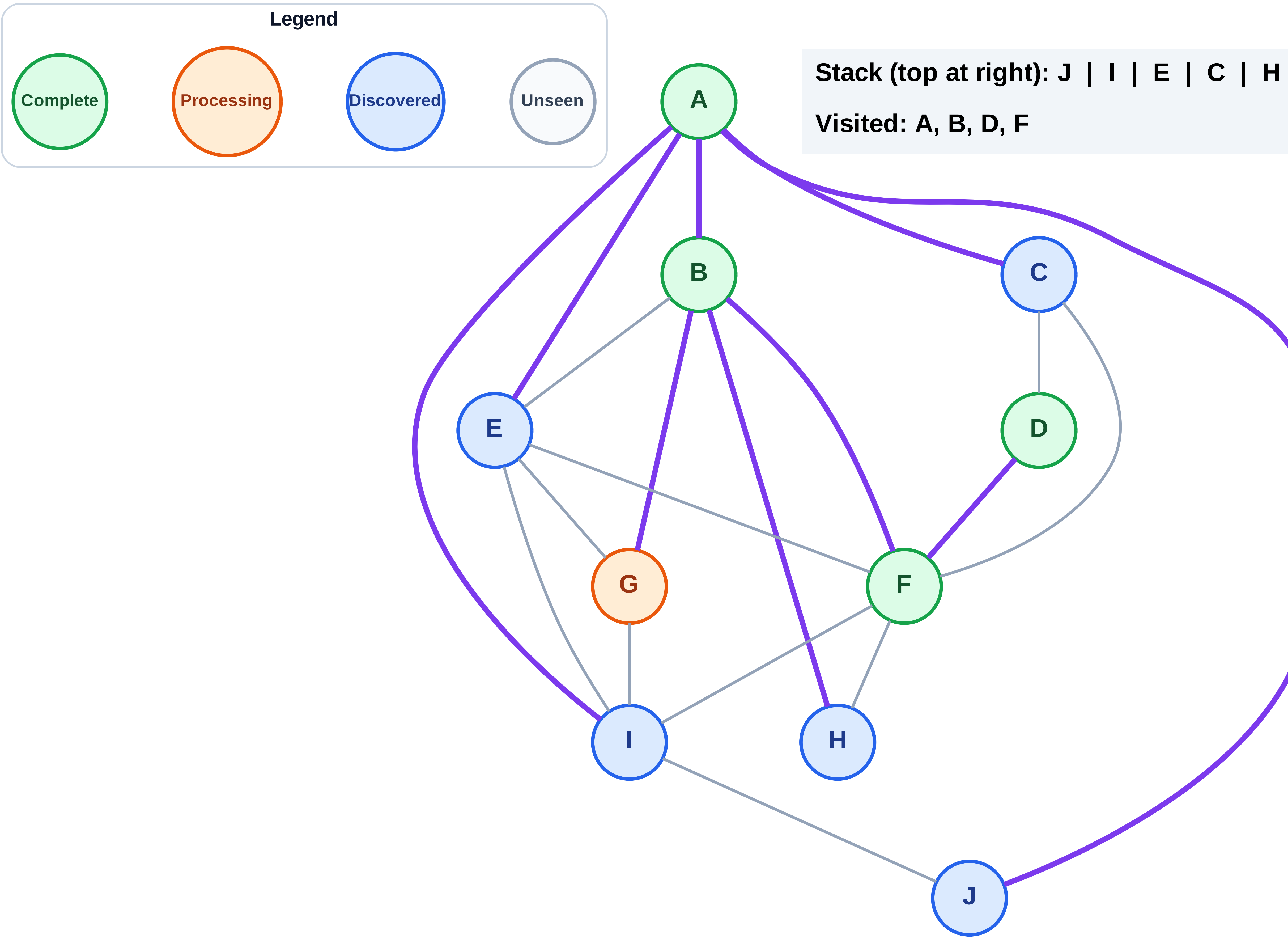
DFS Traversal

Step 9: No new neighbors from D



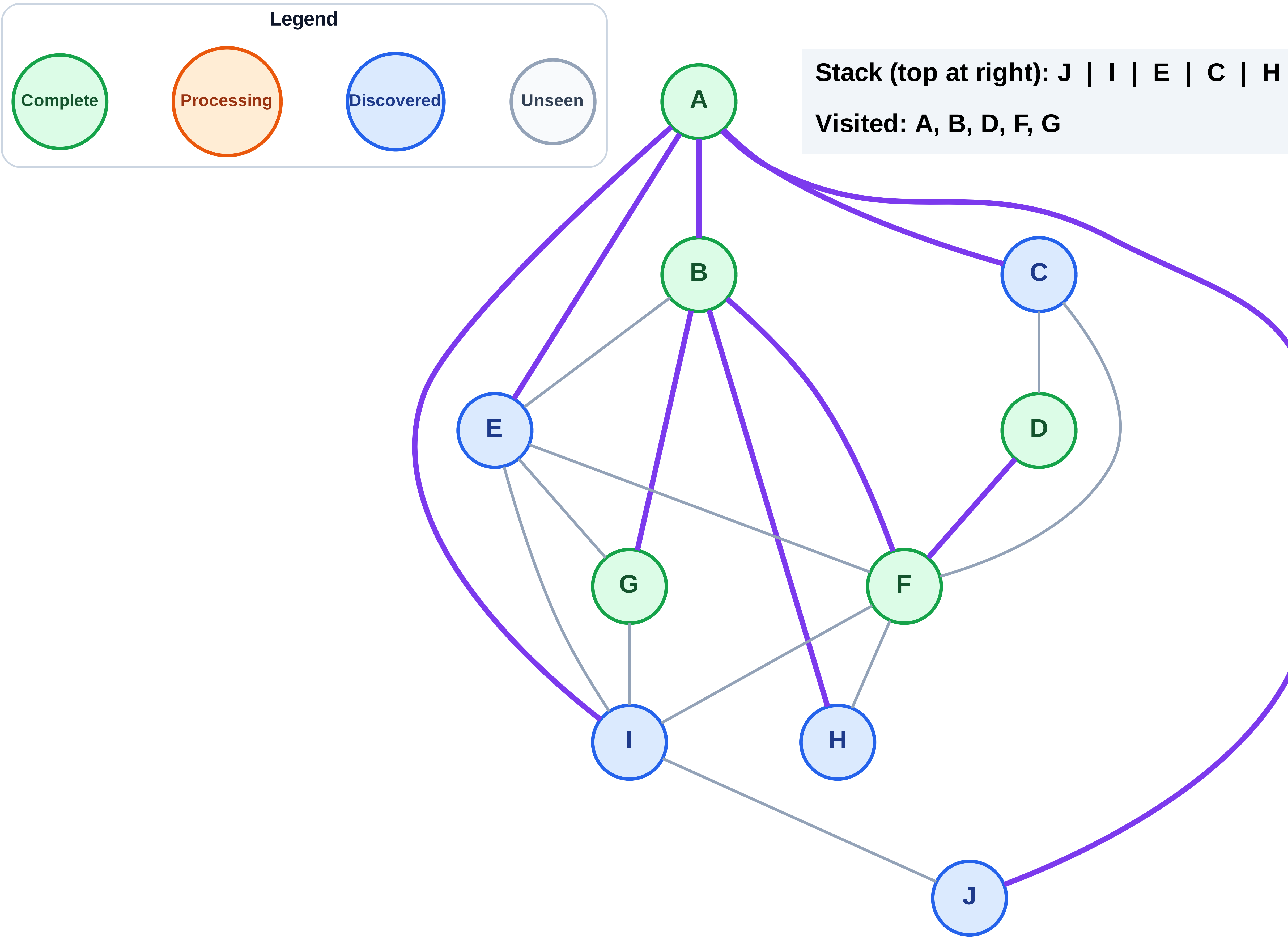
DFS Traversal

Step 10: Process node G



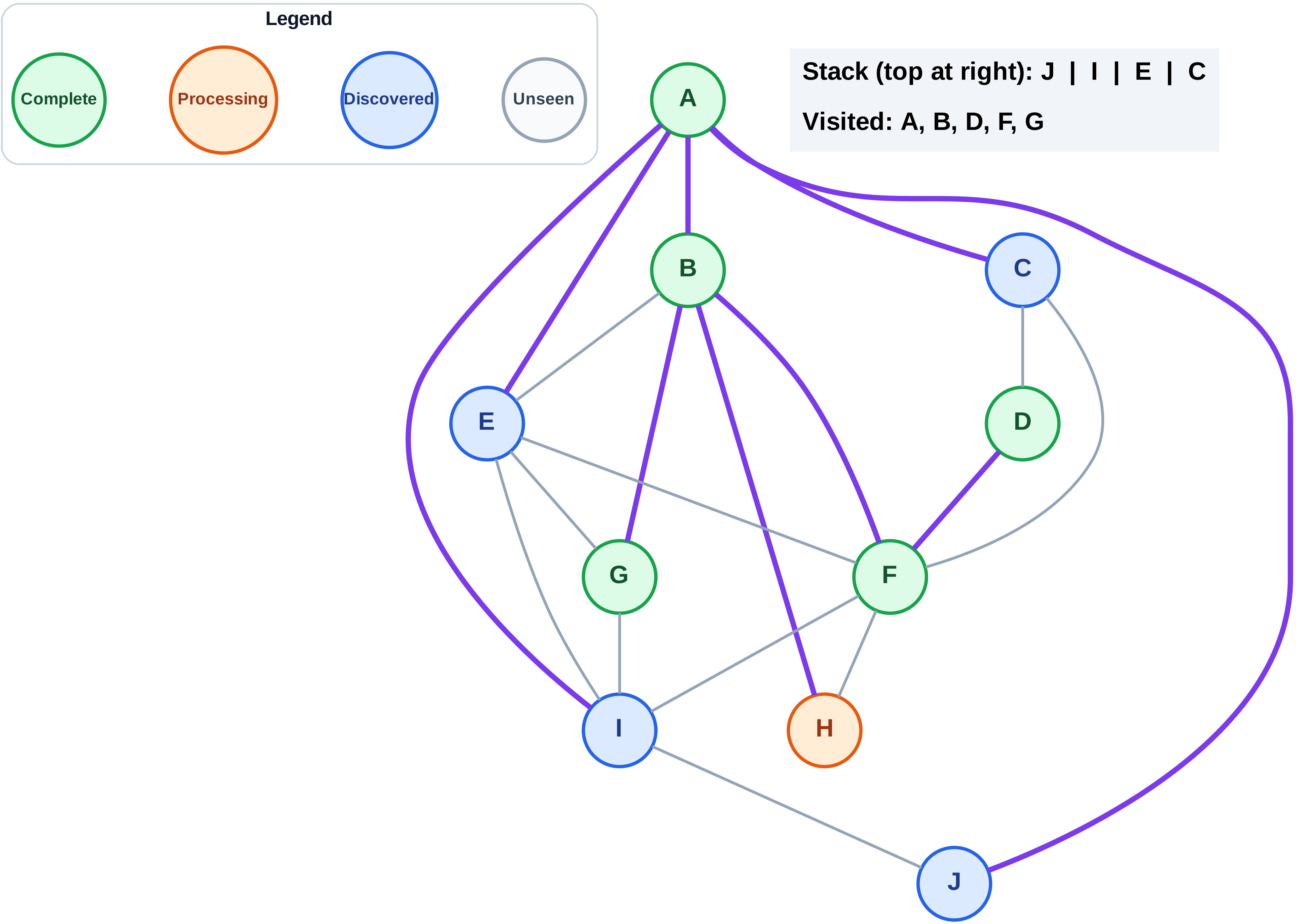
DFS Traversal

Step 11: No new neighbors from G



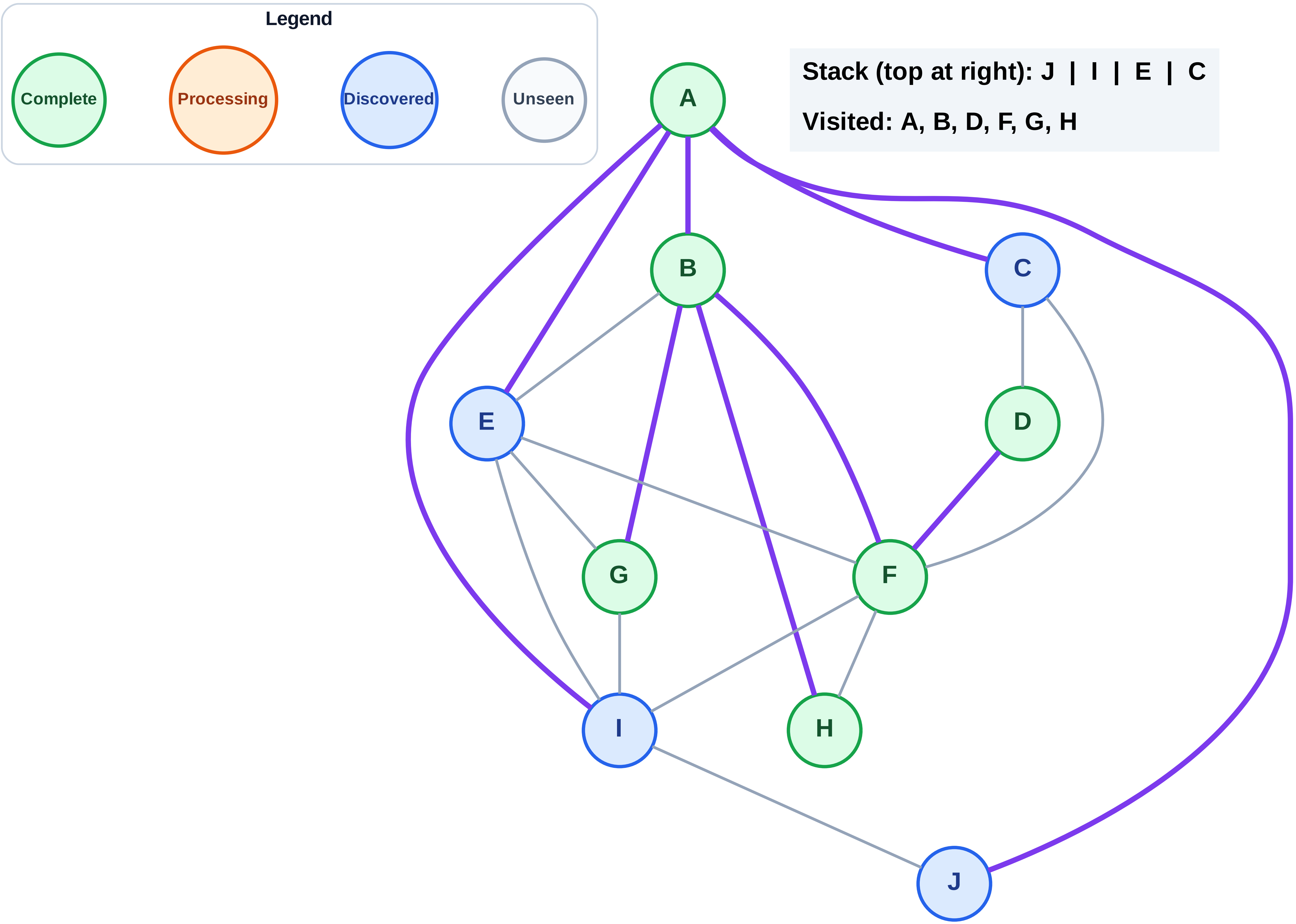
DFS Traversal

Step 12: Process node H



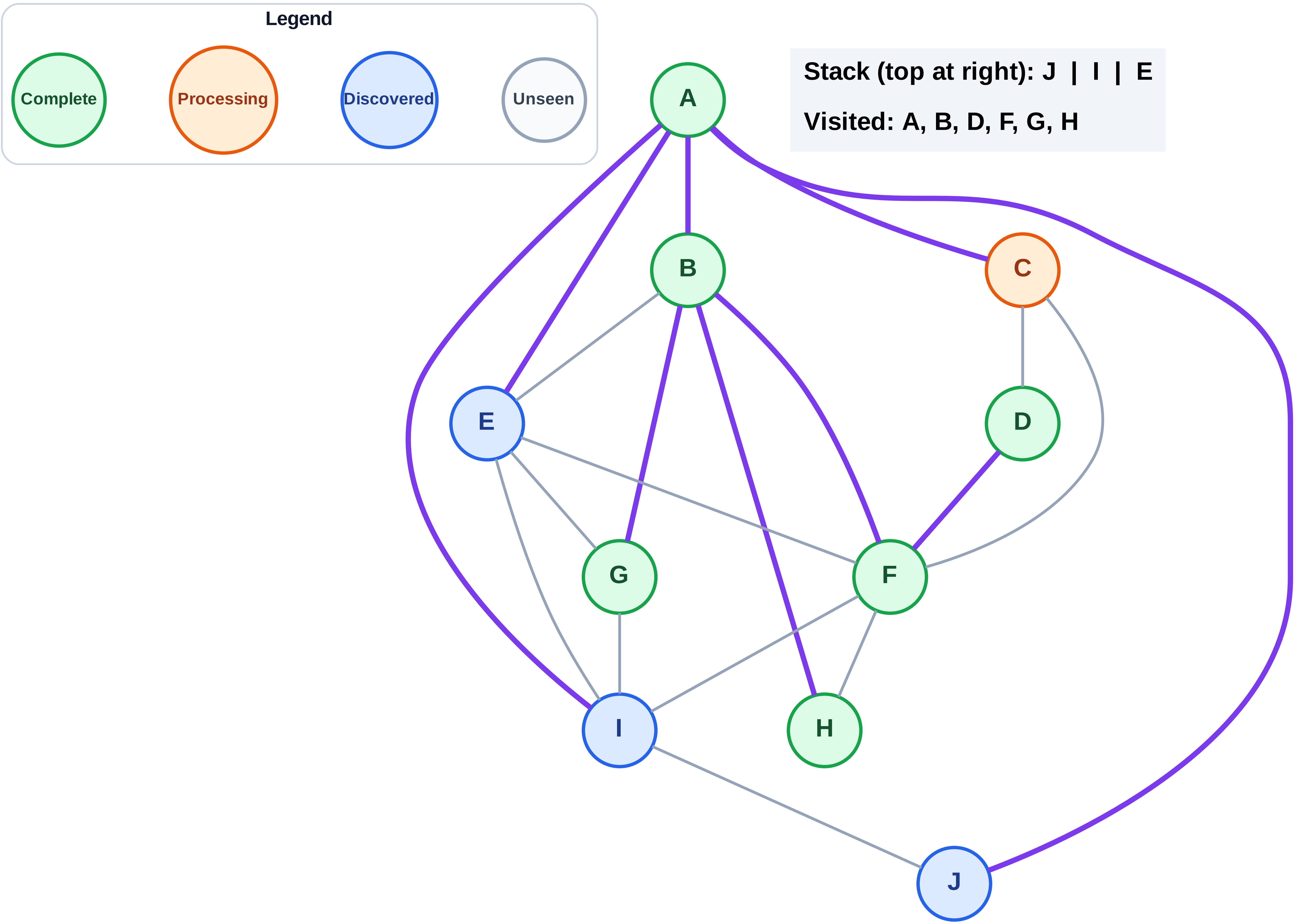
DFS Traversal

Step 13: No new neighbors from H



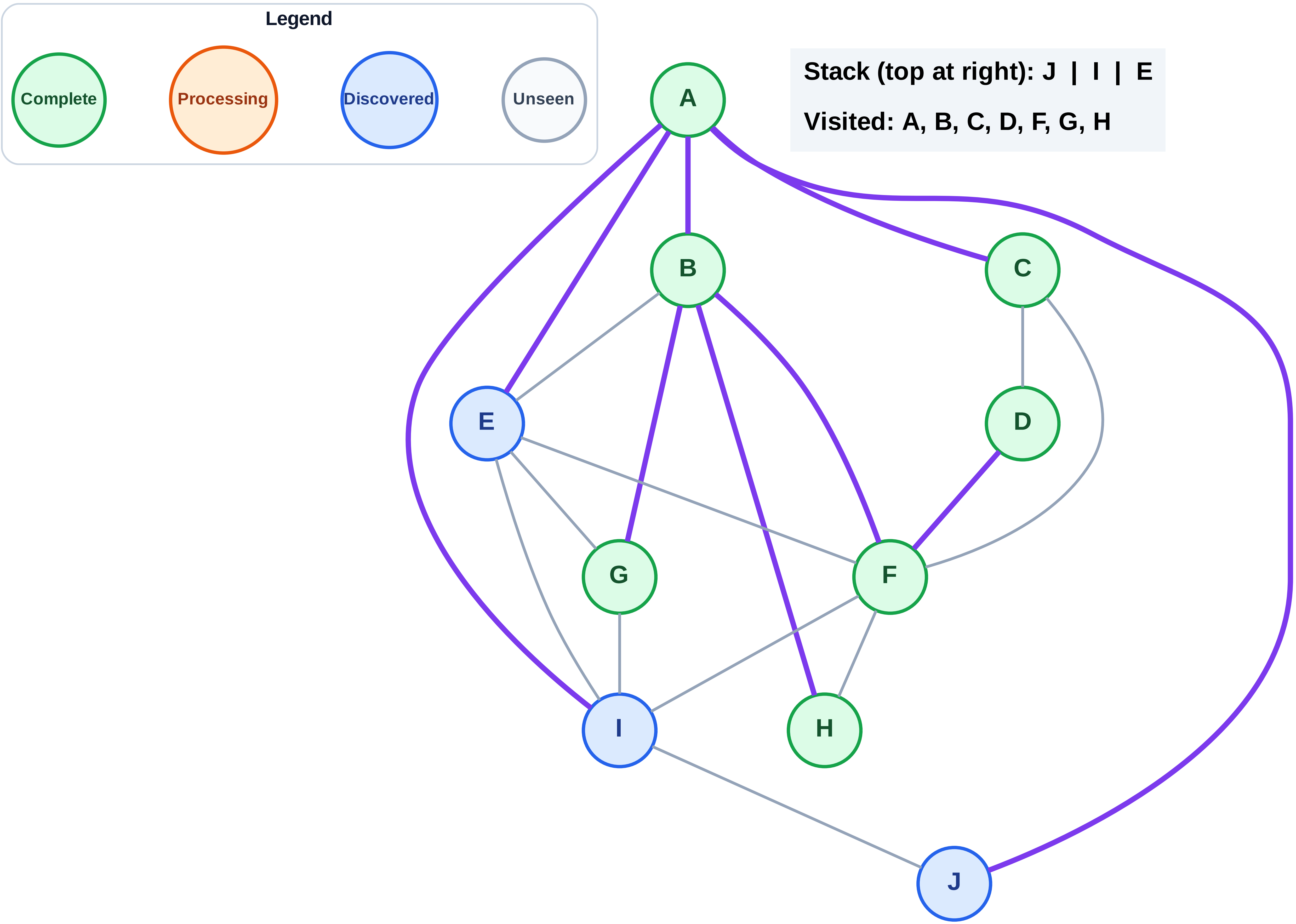
DFS Traversal

Step 14: Process node C



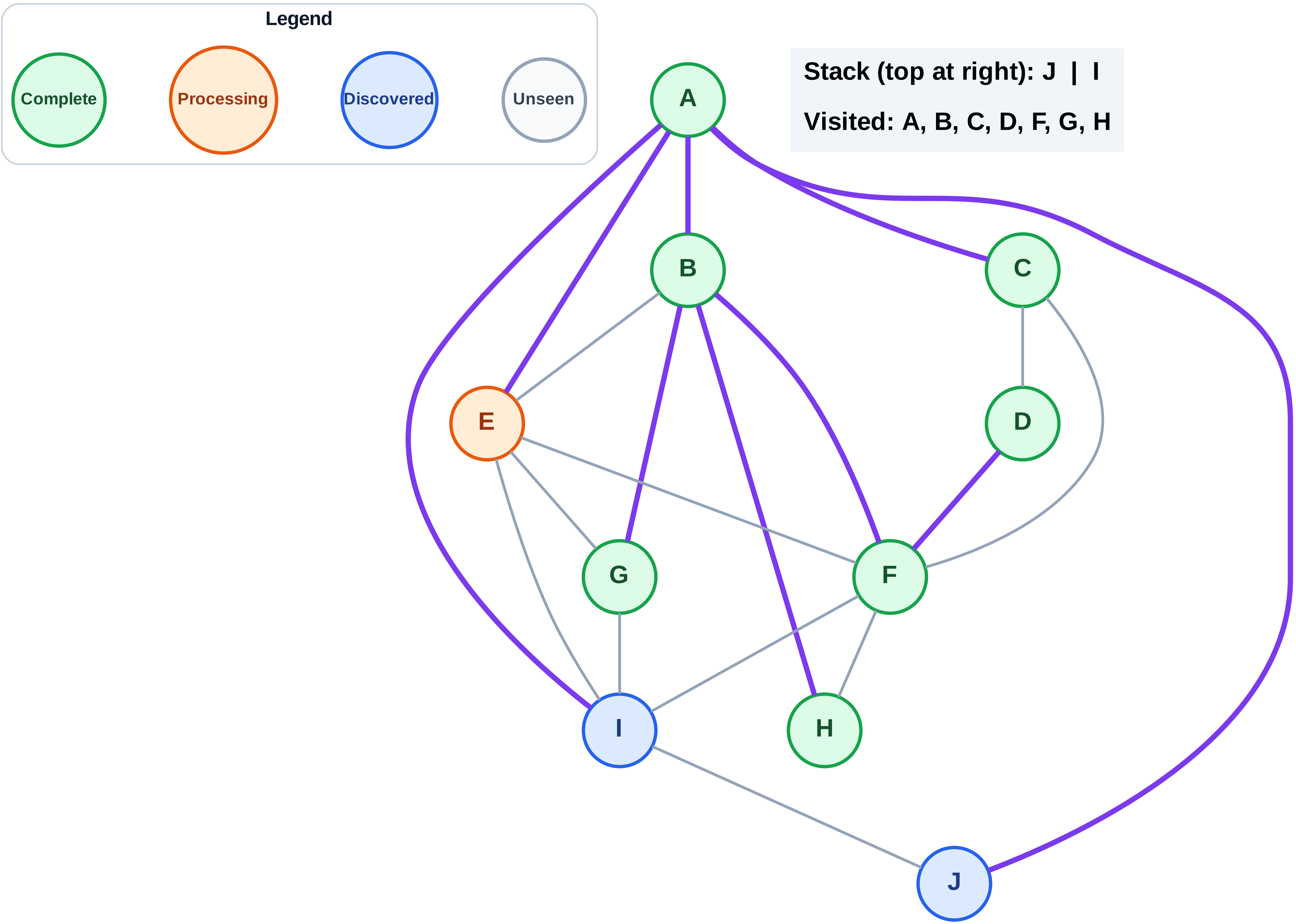
DFS Traversal

Step 15: No new neighbors from C



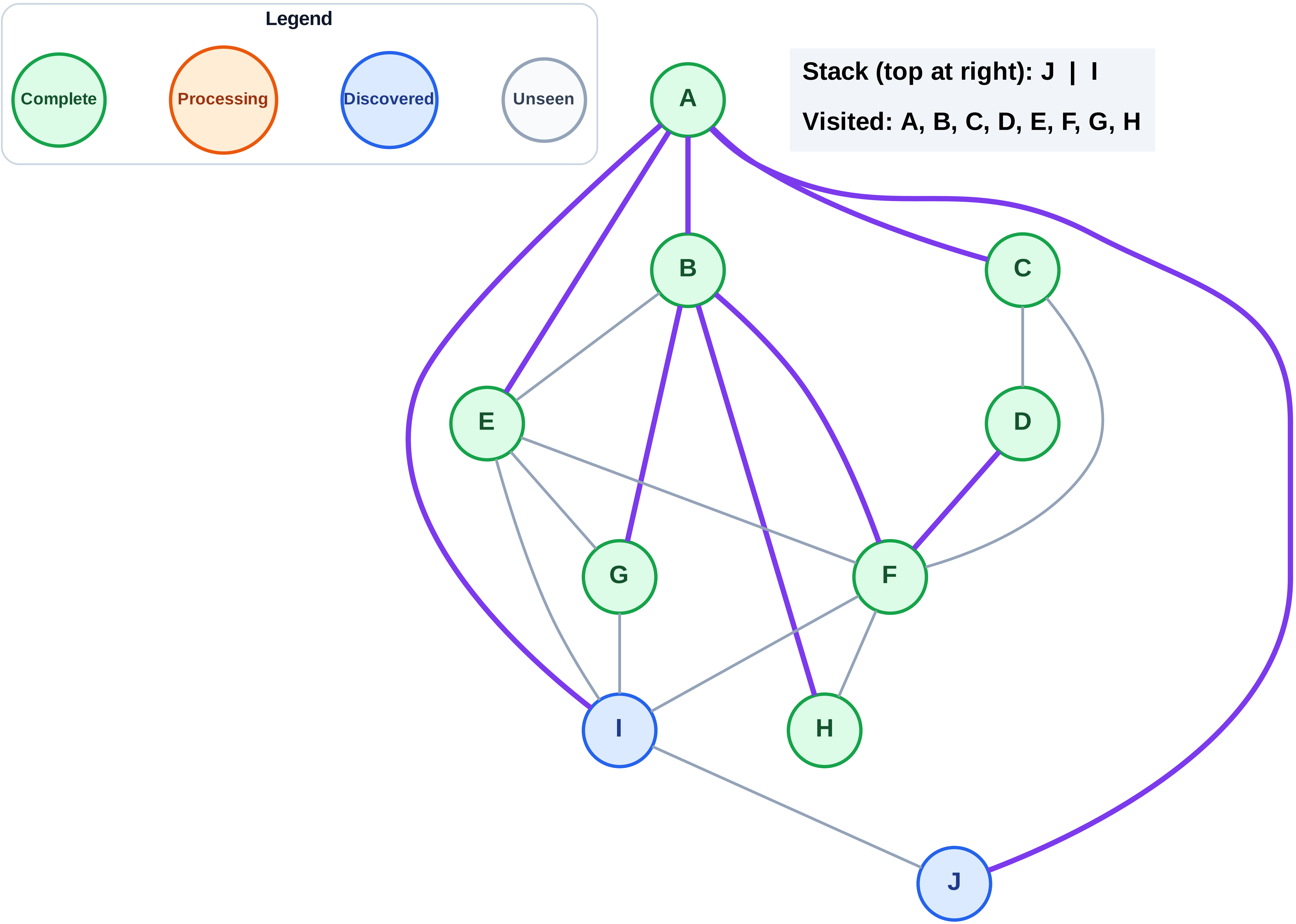
DFS Traversal

Step 16: Process node E



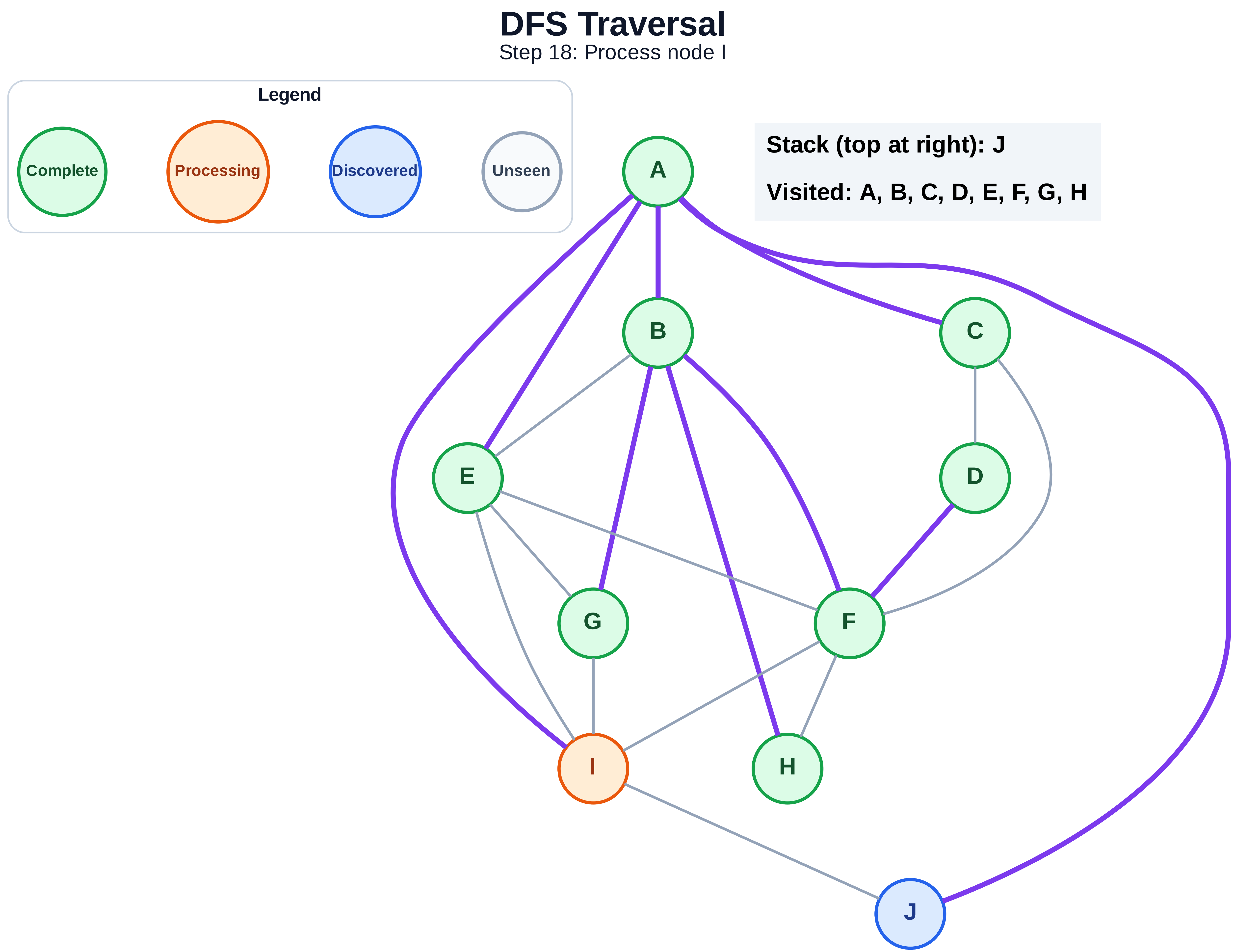
DFS Traversal

Step 17: No new neighbors from E



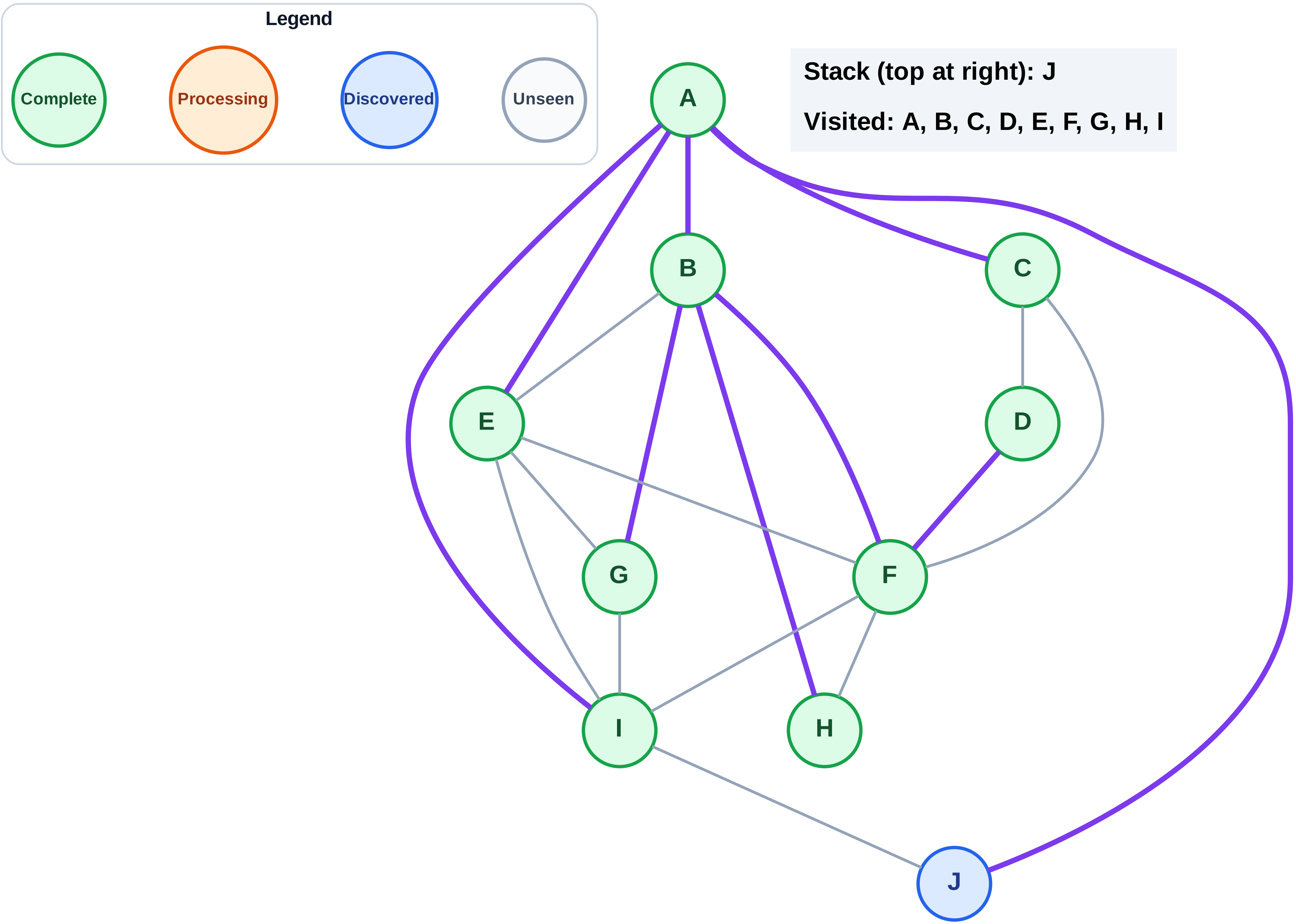
Stack (top at right): J | I

Visited: A, B, C, D, E, F, G, H


DFS Traversal
Step 18: Process node I
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): J
Visited: A, B, C, D, E, F, G, H
A
B
C
D
E
F
G
H
I
J

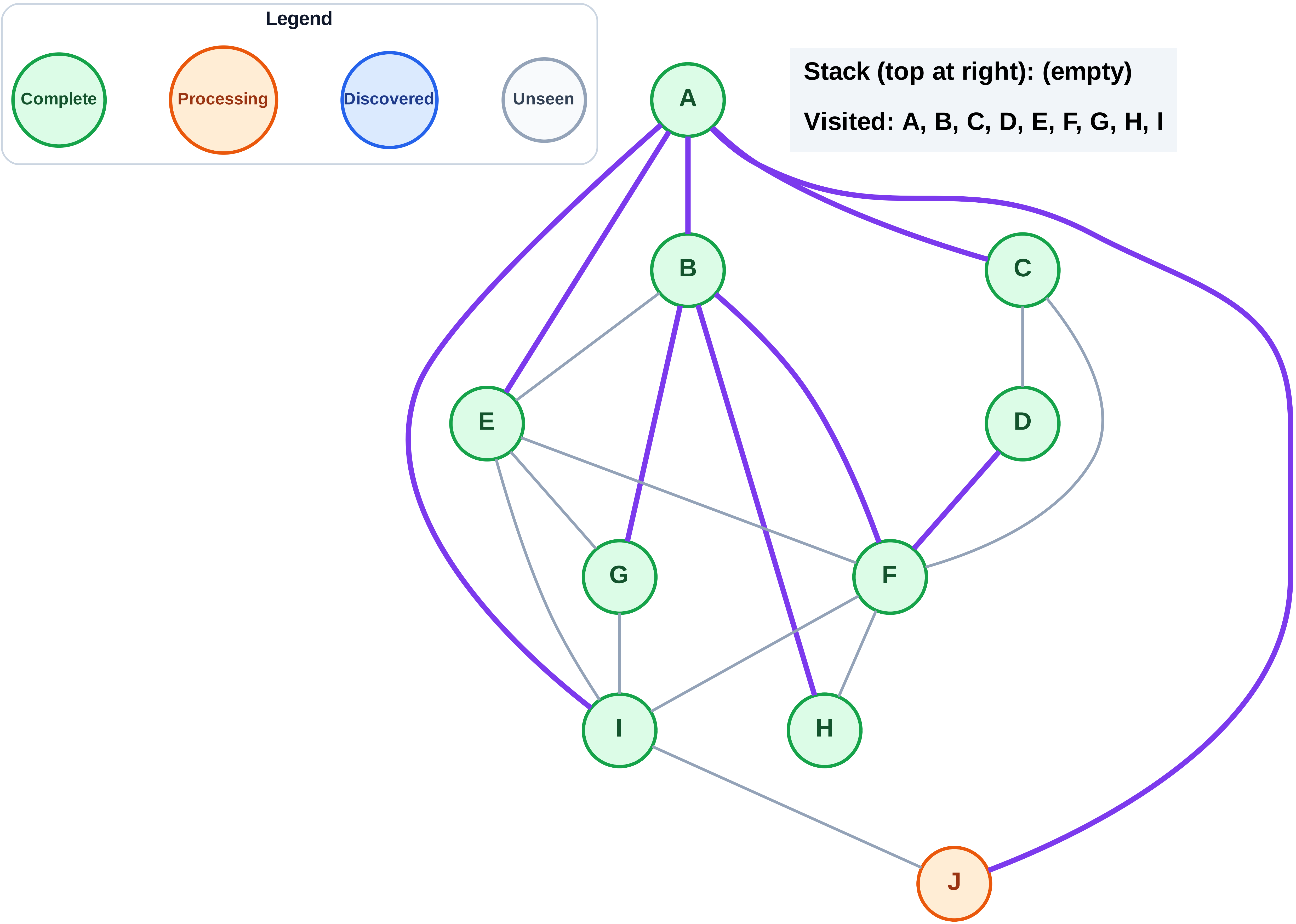
DFS Traversal

Step 19: No new neighbors from I



DFS Traversal

Step 20: Process node J



DFS Traversal

Step 21: No new neighbors from J